第九章 级数

§1. 数项级数

习题. P.182.

1. 讨论下列级数的收敛性，若收敛，试求出级数之和.

(1) $\sum_{n=1}^{\infty}(\sqrt{n+1}-\sqrt{n})$

解: $S_n=\sum_{k=1}^{n}(\sqrt{k+1}-\sqrt{k})=\sqrt{2}-1+\sqrt{3}-\sqrt{2}+\sqrt{4}-\sqrt{3}+\cdots+\sqrt{n}-\sqrt{n-1}+\sqrt{n+1}-\sqrt{n}$

$=\sqrt{n+1}-1.$ $\lim_{n\to\infty}S_n=\lim_{n\to\infty}(\sqrt{n+1}-n)\longrightarrow\infty.$ $\therefore$ 发散

(2) $\sum_{n=1}^{\infty}\frac{1}{\sqrt[n]{n}}$

解: $x_n=\frac{1}{\sqrt[n]{n}}$ $\lim_{n\to\infty}x_n=\lim_{n\to\infty}\frac{1}{\sqrt[n]{n}}=1\not\to 0.$ $\therefore$ 发散.

(3) $\sum_{n=1}^{\infty}\frac{2n-1}{3^n}$

解: $S_n=\sum_{k=1}^{n}\frac{2k-1}{3^k}$ $\frac{1}{3}S_n=\sum_{k=1}^{n}\frac{2k-1}{3^{k-1}}=\sum_{k=0}^{n-1}\frac{2k+1}{3^k}$

两式相减，得: $2S_n=1+\sum_{k=1}^{n-1}\left(\frac{2k+1}{3^k}-\frac{2k-1}{3^k}\right)-\frac{2n-1}{3^n}=1+\sum_{k=1}^{n-1}\frac{2}{3^k}-\frac{2n-1}{3^n}$

$=1+2\cdot\frac{1-(\frac{1}{3})^{n-1}}{1-\frac{1}{3}}-\frac{2n-1}{3^n}=1+3-\frac{1}{3^{n-2}}-\frac{2n-1}{3^n}$

$\therefore S_n=2-\frac{1}{2\times 3^{n-2}}-\frac{2n-1}{2\times 3^n}.$ $\lim_{n\to\infty}S_n=2.$ $\therefore$ 收敛. $\sum_{n=1}^{\infty}\frac{2n-1}{3^n}=2.$

(4) $\sum_{n=1}^{\infty}(\sqrt{n+2}-2\sqrt{n+1}+\sqrt{n})$

解: $S_n=\sum_{k=1}^{n}(\sqrt{k+2}-2\sqrt{k+1}+\sqrt{k})=\sqrt{n+2}-\sqrt{n+1}-\sqrt{2}+1=\frac{1}{\sqrt{n+2}+\sqrt{n+1}}-\sqrt{2}+1.$

$S=\lim_{n\to\infty}S_n=-\sqrt{2}+1.$ $\therefore$ 收敛. $\sum_{n=1}^{\infty}(\sqrt{n+2}-2\sqrt{n+1}+\sqrt{n})=-\sqrt{2}+1.$

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(5) $\sum_{n=1}^{\infty} \frac{1}{(5n-4)(5n+1)}$

解: $x_n = \frac{1}{(5n-4)(5n+1)} = \frac{1}{5}\left(\frac{1}{5n-4} - \frac{1}{5n+1}\right)$

$S_n = \sum_{k=1}^{n} x_k = \frac{1}{5}\left(1 - \frac{1}{6} + \frac{1}{6} - \frac{1}{11} + \frac{1}{11} - \frac{1}{16} + \cdots + \frac{1}{5n-4} - \frac{1}{5n+1}\right) = \frac{1}{5}\left(1 - \frac{1}{5n+1}\right)$.

$S = \lim_{n\to\infty} S_n = \frac{1}{5}$. ∴ 收敛. $\sum_{n=1}^{\infty} \frac{1}{(5n-4)(5n+1)} = \frac{1}{5}$.

(6) $\sum_{n=1}^{\infty} \frac{2n}{3n+1}$.

解: $x_n = \frac{2n}{3n+1}$. $\lim_{n\to\infty} x_n = \lim_{n\to\infty} \frac{2n}{3n+1} = \frac{2}{3} \neq 0$. ∴ 发散.

(7) $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$.

解: $S_n = \sum_{k=1}^{n} \frac{1}{k(k+1)(k+2)} = \frac{1}{2}\sum_{k=1}^{n}\left(\frac{1}{k} - \frac{2}{k+1} + \frac{1}{k+2}\right) = \frac{1}{2}\left(\frac{1}{2} - \frac{1}{n+1} + \frac{1}{n+2}\right)$

$S = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{1}{2}\left(\frac{1}{2} - \frac{1}{n+1} + \frac{1}{n+2}\right) = \frac{1}{4}$. ∴ 收敛.

(8) $\sum_{n=1}^{\infty} \arctan \frac{1}{2n^2}$.

解: $\arctan \frac{1}{2n^2} = \arctan \frac{1}{2n-1} - \arctan \frac{1}{2n+1}$

$S_n = \sum_{k=1}^{n} \arctan \frac{1}{2k^2} = \sum_{k=1}^{n}\left(\arctan \frac{1}{2k-1} - \arctan \frac{1}{2k+1}\right)$

$= \arctan 1 - \arctan \frac{1}{2n+1}$

$S = \lim_{n\to\infty} S_n = \arctan 1 = \frac{\pi}{4}$. ∴ 收敛.

$\arctan \frac{1}{2n-1} = \alpha \Rightarrow \tan\alpha = \frac{1}{2n-1}$

$\arctan \frac{1}{2n+1} = \beta \Rightarrow \tan\beta = \frac{1}{2n+1}$

$\tan(\alpha-\beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha \cdot \tan\beta} = \frac{\frac{1}{2n-1} - \frac{1}{2n+1}}{1 + \frac{1}{4n^2-1}} = \frac{1}{2n^2}$

2. 设抛物线 l_n: $y = nx^2 + \frac{1}{n}$ 和 l_n': $y = (n+1)x^2 + \frac{1}{n+1}$. 以实数轴为对称轴的抛物线 l_n 与 l_n' 所围成的平面图形的面积为 a_n ($n = 1, 2, \cdots$). (1) 求抛物线 l_n 与 l_n' 所围成的平面图形的面积 S_n;

(2) 求级数 $\sum_{n=1}^{\infty} \frac{S_n}{a_n}$ 的和

解: $\begin{cases} y = nx^2 + \frac{1}{n} \\ y = (n+1)x^2 + \frac{1}{n+1} \end{cases}$ $\Rightarrow$ l_n 与 l_n' 的交横坐标 $a_n = \frac{1}{\sqrt{n(n+1)}}$ (正对值).

(1) $S_n = 2\int_0^{a_n} \left[\left(nx^2 + \frac{1}{n}\right) - \left((n+1)x^2 + \frac{1}{n+1}\right) \right] dx = 2\int_0^{a_n} \left(\frac{1}{n(n+1)} - x^2 \right) dx$

$= 2\left[\frac{1}{n(n+1)} x - \frac{1}{3}x^3 \right]_0^{a_n} = 2\left[\frac{1}{n(n+1)} \cdot a_n - \frac{1}{3}a_n^3 \right] = \frac{4}{3}a_n^3$.

(2) $\sum_{n=1}^{\infty} \frac{S_n}{a_n} = \sum_{n=1}^{\infty} \frac{4}{3}a_n^2 = \frac{4}{3}\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{4}{3}\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \frac{4}{3}$.

3. 利用 Cauchy 收敛原理证明下述级数发散:

(1) $1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} + \frac{1}{8} - \frac{1}{9} + \cdots$;

证明: 设通项为 x_n. 则:

$x_{3n+1} + x_{3n+2} + \cdots + x_{6n} > x_{3n+1} + x_{3n+4} + \cdots + x_{6n-2}$

$= \frac{1}{3n+1} + \frac{1}{3n+4} + \cdots + \frac{1}{6n-2} > \frac{n}{6n-2} > \frac{1}{6}$.

($x_{3n+2} + x_{3n+3} = \frac{1}{3n+2} - \frac{1}{3n+3} = \frac{1}{(3n+2)(3n+3)} > 0$)

($1 + \left(\frac{1}{2} - \frac{1}{3}\right) + \frac{1}{4} + \left(\frac{1}{5} - \frac{1}{6}\right) + \frac{1}{7} + \left(\frac{1}{8} - \frac{1}{9}\right) + \cdots > 1 + \frac{1}{4} + \frac{1}{7} + \cdots$, 各括号 > 0)

由 Cauchy 原理 ($\exists \varepsilon > 0$, $\forall N$, $\exists n > m > N$: $|S_n - S_m| > \varepsilon$).

取 $\varepsilon = \frac{1}{6}$. $\forall N$. 取 $m = 3N$. $n = 6N$. 则:

$|S_n - S_m| = |x_{3N+1} + x_{3N+2} + \cdots + x_{3N}| > \frac{1}{6} = \varepsilon$.

$\therefore$ 原级数发散.

(2) $1 - \frac{1}{2} + \frac{1}{3} + \frac{1}{4} - \frac{1}{5} + \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \frac{1}{9} + \cdots$

证明: 设通项为 x_n. 则:

$x_{3n+1} + x_{3n+2} + \cdots + x_{6n} > x_{3n+3} + x_{3n+6} + \cdots + x_{6n}$

$= \frac{1}{3n+3} + \frac{1}{3n+6} + \cdots + \frac{1}{6n} > \frac{n}{6n} = \frac{1}{6}$.

($x_{3n+1} + x_{3n+2} = \frac{1}{3n+1} - \frac{1}{3n+2} = \frac{1}{(3n+1)(3n+2)} > 0$)

($\left(1 - \frac{1}{2}\right) + \frac{1}{3} + \left(\frac{1}{4} - \frac{1}{5}\right) + \frac{1}{6} + \left(\frac{1}{7} - \frac{1}{8}\right) + \frac{1}{9} + \cdots$, 各括号 > 0)

$\therefore$ 发散 (同上).

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4. 讨论下列正项级数的收敛性：

(1) $\sum_{n=1}^{\infty} \frac{1}{n!}$

解：$\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = \lim_{n\to\infty} \frac{1}{(n+1)!} \cdot \frac{n!}{1} = \lim_{n\to\infty} \frac{1}{n+1} = 0 < 1$. ∴ 收敛.

(2) $\sum_{n=1}^{\infty} \frac{1}{\sqrt[n]{n}}$

解：$\lim_{n\to\infty} x_n = \lim_{n\to\infty} \frac{1}{\sqrt[n]{n}} = 1 \not\to 0$ ∴ 发散.

(3) $\sum_{n=1}^{\infty} \left(1-\cos\frac{\pi}{n}\right)$.

解：$1-\cos\frac{\pi}{n} = 2\sin^2\frac{\pi}{2n} \sim 2\cdot\left(\frac{\pi}{2n}\right)^2 = \frac{\pi^2}{2n^2}$. $(n\to\infty)$.

∵ $\sum_{n=1}^{\infty} \frac{\pi^2}{2n^2}$ 收敛. ∴ $\sum_{n=1}^{\infty} \left(1-\cos\frac{\pi}{n}\right)$ 收敛.

(4) $\sum_{n=1}^{\infty} \frac{\ln n}{n^2}$.

解：$\frac{\ln n}{n^2} < \frac{1}{n\sqrt{n}}$. ∵ $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}}$ 收敛 (p 级数，$p>1$ 时收敛). ∴ $\sum_{n=1}^{\infty} \frac{\ln n}{n^2}$ 收敛.

(5) $\sum_{n=1}^{\infty} \frac{4n}{n^4+1}$.

解：$\frac{4n}{n^4+1} \sim \frac{4}{n^3}$ $(n\to\infty)$. ∵ $\sum_{n=1}^{\infty} \frac{4}{n^3}$ 收敛. ∴ $\sum_{n=1}^{\infty} \frac{4n}{n^4+1}$ 收敛.

(6) $\sum_{n=1}^{\infty} \frac{1}{\ln^n(n+2)}$.

解：$\lim_{n\to\infty} \sqrt[n]{x_n} = \lim_{n\to\infty} \frac{1}{\ln(n+2)} = 0 < 1$. ∴ 收敛.

(7) $\sum_{n=1}^{\infty} \left(\sqrt[n]{n}-1\right)$

解：当 $n \geq 3$ 时，$\sqrt[n]{n}-1 > e^{\frac{1}{n}}-1 \sim \frac{1}{n}$ $(n\to\infty)$

∵ $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散. ∴ $\sum_{n=1}^{\infty} \left(\sqrt[n]{n}-1\right)$ 发散.

(8) $\sum_{n=1}^{\infty}\left(\frac{n}{3n+1}\right)^n$

解: $\lim_{n\to\infty}\sqrt[n]{x_n}=\lim_{n\to\infty}\frac{n}{3n+1}=\frac{1}{3}<1$. ∴ 收敛.

(9) $\sum_{n=1}^{\infty}\frac{n^2}{2^n}$

解: $\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{(n+1)^2}{2^{n+1}}\cdot\frac{2^n}{n^2}=\frac{1}{2}\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^2=\frac{1}{2}<1$. ∴ 收敛.

(10) $\sum_{n=1}^{\infty}\frac{[2+(-1)^n]^n}{2^{2n+1}}$

解: $x_n=\frac{[2+(-1)^n]^n}{2^{2n+1}}$. $\lim_{n\to\infty}\sqrt[n]{x_n}=\lim_{n\to\infty}\frac{2+(-1)^n}{4}\le\frac{3}{4}<1$. ∴ 收敛.

(11) $\sum_{n=1}^{\infty}n^2e^{-n}$

解: $\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{(n+1)^2}{e^{n+1}}\cdot\frac{e^n}{n^2}=\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^2\cdot\frac{1}{e}=\frac{1}{e}<1$. ∴ 收敛.

(12) $\sum_{n=1}^{\infty}n\tan\frac{\pi}{2^{n+1}}$.

解: $\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{n+1}{n}\cdot\frac{\tan\frac{\pi}{2^{n+1}}\cdot\frac{1}{2}}{\tan\frac{\pi}{2^{n+1}}}=\lim_{n\to\infty}\frac{n+1}{n}\cdot\frac{1-\tan^2\frac{\pi}{2^{n+1}}}{2}=\frac{1}{2}<1$. ∴ 收敛.

(13) $\sum_{n=1}^{\infty}\left(\sqrt{n^2+1}-\sqrt{n^2-1}\right)$

解: $\sqrt{n^2+1}-\sqrt{n^2-1}=\frac{2}{\sqrt{n^2+1}+\sqrt{n^2-1}}\sim\frac{1}{n}\ (n\to\infty)$.

∵ $\sum_{n=1}^{\infty}\frac{1}{n}$ 发散. ∴ $\sum_{n=1}^{\infty}\left(\sqrt{n^2+1}-\sqrt{n^2-1}\right)$ 发散.

(14) $\sum_{n=1}^{\infty}\left(2n-\sqrt{n^2+1}-\sqrt{n^2-1}\right)$

解: $2n-\sqrt{n^2+1}-\sqrt{n^2-1}=\frac{2\left(n^2-\sqrt{(n^2+1)(n^2-1)}\right)}{2n+\sqrt{n^2+1}+\sqrt{n^2-1}}=\frac{2}{2n+\sqrt{n^2+1}+\sqrt{n^2-1}}\cdot\frac{1}{n^2+\sqrt{(n^2+1)(n^2-1)}}$

$\sim\frac{2}{2n+n+n}\cdot\frac{1}{n^2+n^2}=\frac{1}{4n^3}\ (n\to\infty)$. ∵ $\sum_{n=1}^{\infty}\frac{1}{4n^3}$ 收敛. ∴ 原级数收敛.

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(15) $\sum_{n=2}^{\infty} \ln\frac{n^2+1}{n^2-1}$

解: $\ln\frac{n^2+1}{n^2-1} \sim \ln\left(1+\frac{2}{n^2-1}\right) \sim \frac{2}{n^2-1} \sim \frac{2}{n^2}$ $(n\to\infty)$

$\because \sum_{n=1}^{\infty}\frac{2}{n^2}$ 收敛 $\therefore \sum_{n=2}^{\infty}\ln\frac{n^2+1}{n^2-1}$ 收敛.

(16) $\sum_{n=1}^{\infty}\left(e^{\frac{1}{n^2}}-\cos\frac{\pi}{n}\right)$.

解: $e^{\frac{1}{n^2}}-\cos\frac{\pi}{n} = \left(e^{\frac{1}{n^2}}-1\right)+\left(1-\cos\frac{\pi}{n}\right) \sim \frac{1}{n^2}+2\sin^2\frac{\pi}{2n} = \frac{1}{n^2}+2\cdot\left(\frac{\pi}{2n}\right)^2 = \frac{2+\pi^2}{2n^2}$.

$\because \sum_{n=1}^{\infty}\frac{2+\pi^2}{2n^2}$ 收敛 $\therefore \sum_{n=1}^{\infty}\left(e^{\frac{1}{n^2}}-\cos\frac{\pi}{n}\right)$ 收敛.

(17) $\sum_{n=2}^{\infty}\frac{1}{n\cdot\ln n\cdot\ln\ln n}$

解: $f(x)=\frac{1}{x\cdot\ln x\cdot\ln\ln x}$ $\int_2^{+\infty}f(x)\,dx = \int_2^{+\infty}\frac{1}{x\cdot\ln x\cdot\ln\ln x}\,dx = \ln|\ln(\ln x)|\Big|_2^{+\infty}$

$\because \int_2^{+\infty}f(x)\,dx$ 发散. $\therefore \sum_{n=2}^{\infty}\frac{1}{n\cdot\ln n\cdot\ln\ln n}$ 发散.

(18) $\sum_{n=2}^{\infty}\frac{1}{n\cdot(\ln n)^{1+p}\cdot\ln\ln n}$ $(p>0)$.

解: 当 $n>e^e$ 时, $\ln n>e$. $\ln\ln n>1$. $\therefore \frac{1}{n\cdot(\ln n)^{1+p}\cdot\ln\ln n} < \frac{1}{n\cdot(\ln n)^{1+p}}$.

取 $f(x)=\frac{1}{x\cdot(\ln x)^{1+p}}$ $\int_2^{+\infty}f(x)\,dx = \int_2^{+\infty}\frac{1}{x\cdot(\ln x)^{1+p}}\,dx$

$= \int_2^{+\infty}\frac{1}{(\ln x)^{1+p}}\,d(\ln x) = -\frac{1}{p}(\ln x)^{-p}\Big|_2^{+\infty} = \frac{1}{p}\cdot\frac{1}{\ln^p 2}$.

即 $\sum_{n=2}^{\infty}\frac{1}{n\cdot(\ln n)^{1+p}}$ 收敛. $\therefore \sum_{n=2}^{\infty}\frac{1}{n\cdot(\ln n)^{1+p}\cdot\ln\ln n}$ 收敛. $(p>0)$.

5. 利用级数收敛的必要条件, 证明:

(1) $\lim_{n\to\infty}\frac{n^n}{(n!)^2}=0$.

证明：设 $x_n=\frac{n^n}{(n!)^2}$，$\lim\limits_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim\limits_{n\to\infty}\frac{(n+1)^{n+1}}{(n+1)!\cdot(n+1)!}\cdot\frac{n!\cdot n!}{n^n}$

$=\lim\limits_{n\to\infty}\frac{1}{n+1}\cdot\left(1+\frac{1}{n}\right)^n=0<1$，$\therefore\sum\limits_{n=1}^{\infty}x_n$ 收敛。

则通项 x_n 极限必趋于0。$\therefore\lim\limits_{n\to\infty}x_n=\lim\limits_{n\to\infty}\frac{n^n}{(n!)^2}=0$。

(2) $\lim\limits_{n\to\infty}\frac{(2n)!}{2^{n(n+1)}}=0$。

证明：设 $x_n=\frac{(2n)!}{2^{n(n+1)}}$，$\lim\limits_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim\limits_{n\to\infty}\frac{(2n+2)!}{2^{(n+1)(n+2)}}\cdot\frac{2^{n(n+1)}}{(2n)!}$

$=\lim\limits_{n\to\infty}\frac{(2n+2)(2n+1)}{2^{2n+2}}=0<1$，$\therefore\sum\limits_{n=1}^{\infty}x_n$ 收敛。

则通项 x_n 极限必趋于0。$\therefore\lim\limits_{n\to\infty}x_n=\lim\limits_{n\to\infty}\frac{(2n)!}{2^{n(n+1)}}=0$。

6. 讨论下列级数的收敛性：

(1) $\sum\limits_{n=1}^{\infty}\int_0^{\frac{1}{n}}\sqrt{\frac{x}{1-x}}\,dx$。

解：当 $n\geq 2$ 时，$\left(x<\frac{1}{n}\leq\frac{1}{2}\right)$

$\int_0^{\frac{1}{n}}\sqrt{\frac{x}{1-x}}\,dx<\int_0^{\frac{1}{n}}\sqrt{2x}\,dx=\frac{1}{2}\int_0^{\frac{1}{n}}(2x)^{\frac{1}{2}}\,d(2x)$

$=\frac{1}{2}\cdot\frac{2}{3}(2x)^{\frac{3}{2}}\Big|_0^{\frac{1}{n}}=\frac{1}{3}\cdot\frac{2}{n}\sqrt{\frac{2}{n}}<\frac{1}{n\sqrt{n}}$

$\because\sum\limits_{n=1}^{\infty}\frac{1}{n\sqrt{n}}$ 收敛。（p 级数：$p=\frac{3}{2}$）。$\therefore\sum\limits_{n=1}^{\infty}\int_0^{\frac{1}{n}}\sqrt{\frac{x}{1-x}}\,dx$ 收敛。

(2) $\sum\limits_{n=1}^{\infty}\int_{n\pi}^{2n\pi}\frac{\sin^2x}{x^2}\,dx$。

解：$\int_{n\pi}^{2n\pi}\frac{\sin^2x}{x^2}\,dx>\int_{n\pi}^{2n\pi}\frac{\sin^2x}{(2n\pi)^2}\,dx$ $(n\pi<x<2n\pi)\Rightarrow\frac{1}{x^2}>\frac{1}{(2n\pi)^2}$

$=\frac{1}{4n^2\pi^2}\int_{n\pi}^{2n\pi}\sin^2x\,dx=\frac{1}{8n\pi}$。

$\because\sum\limits_{n=1}^{\infty}\frac{1}{8n\pi}$ 发散。$\therefore\sum\limits_{n=1}^{\infty}\int_{n\pi}^{2n\pi}\frac{\sin^2x}{x^2}\,dx$ 发散。

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(3) $\sum_{n=1}^{\infty} \int_0^{\frac{1}{n}} \ln(1+x)\,dx$

解: $\int_0^{\frac{1}{n}} \ln(1+x)\,dx < \int_0^{\frac{1}{n}} x\,dx = \frac{1}{2n^2}$ $(\ln(1+x) \le x,\ 0<x<\frac{1}{n})$.

$\because \sum_{n=1}^{\infty} \frac{1}{2n^2}$ 收敛. $\therefore \sum_{n=1}^{\infty} \int_0^{\frac{1}{n}} \ln(1+x)\,dx$ 收敛.

7. 设正项级数 $\sum_{n=1}^{\infty} x_n$ 收敛, 证明 $\sum_{n=1}^{\infty} x_n^2$ 也收敛, 反之如何?

证明: 设正项级数 $\sum_{n=1}^{\infty} x_n$ 收敛, 则 $\lim_{n\to\infty} x_n = 0$. 即当 n 充分大时,

有 $0 \le x_n < 1$. 即 $x_n^2 \le x_n$ $\therefore \sum_{n=1}^{\infty} x_n^2$ 收敛.

反之, 当 $\sum_{n=1}^{\infty} x_n^2$ 收敛, 则 $\sum_{n=1}^{\infty} x_n$ 不一定收敛.

如: $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 收敛. 但 $\sum_{n=1}^{\infty} \frac{1}{n}$ 不收敛.

8. 设正项级数 $\sum_{n=1}^{\infty} x_n$ 收敛, 证明当 $p>\frac{1}{2}$ 时, 级数 $\sum_{n=1}^{\infty} \frac{\sqrt{x_n}}{n^p}$ 也收敛; 又问当 $0<p\le\frac{1}{2}$ 时, 结论是否仍然成立?

证明: 设正项级数 $\sum_{n=1}^{\infty} x_n$ 收敛, 当 $p>\frac{1}{2}$ 时, 由 $\frac{\sqrt{x_n}}{n^p} \le \frac{1}{2}\left(x_n + \frac{1}{n^{2p}}\right)$ $\left(ab \le \frac{1}{2}(a^2+b^2)\right)$.

而 $p>\frac{1}{2}$ 时, $\sum_{n=1}^{\infty} x_n$ 与 $\sum_{n=1}^{\infty} \frac{1}{n^{2p}}$ 均收敛. $\therefore \sum_{n=1}^{\infty} \frac{\sqrt{x_n}}{n^p}$ 也收敛.

而当 $0<p\le\frac{1}{2}$ 时, $\sum_{n=1}^{\infty} \frac{\sqrt{x_n}}{n^p}$ 不一定收敛.

如: $x_n = \frac{1}{n \ln^2 n}$. 则有 $\sum_{n=1}^{\infty} x_n$ 收敛 但 $\sum_{n=1}^{\infty} \frac{\sqrt{x_n}}{n^p} = \sum_{n=1}^{\infty} \frac{1}{n^{(\frac{1}{2}+p)} \ln n}$ 发散.

9. 设 $x_n>0$, $\frac{x_{n+1}}{x_n} > 1-\frac{1}{n}$ $(n=1,2,\cdots)$. 证明 $\sum_{n=1}^{\infty} x_n$ 发散.

证明: 由 $x_n>0$, $\frac{x_{n+1}}{x_n} > 1-\frac{1}{n}$. 可知: $(n+1)\,x_{n+1} < n\,x_n$.

即数列 $\{n x_{n+1}\}$ 单调增加, 存在 $\alpha>0$, 使得 $n x_{n+1} \ge \alpha$, 即 $x_{n+1} \ge \frac{\alpha}{n}$.

$\because \sum_{n=1}^{\infty} \frac{\alpha}{n}$ 发散. $\therefore \sum_{n=1}^{\infty} x_n$ 发散.

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10. 讨论下列级数的收敛性（包括条件收敛与绝对收敛）

(1) $\sum_{n=1}^{\infty}(-1)^{n+1}\sin\frac{x}{n}$

解：当 $x=0$ 时，$x_n=0$. 级数绝对收敛.

当 $x\neq 0$ 时，若 $n>\frac{2|x|}{\pi}$，$\sum_{n=1}^{\infty}(-1)^{n+1}\sin\frac{x}{n}$ 是交错级数.

且 $\left|\sin\frac{x}{n}\right|$ 单调减少趋于0. $\therefore \sum_{n=1}^{\infty}(-1)^{n+1}\sin\frac{x}{n}$ 收敛.

$\left|(-1)^{n+1}\sin\frac{x}{n}\right| = \left|\sin\frac{x}{n}\right| \sim \frac{|x|}{n}$ $(n\to\infty)$. $\because \sum_{n=1}^{\infty}\frac{|x|}{n}$ 发散.

$\therefore \sum_{n=1}^{\infty}\left|(-1)^{n+1}\sin\frac{x}{n}\right|$ 发散. 即原级数条件收敛.

(2) $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{2^n\sin^{2n}x}{n}$

解：当 $x\in(k\pi-\frac{\pi}{4},\ k\pi+\frac{\pi}{4})$ 时，由于 $\left|(-1)^{n+1}\frac{2^n\sin^{2n}x}{n}\right| = \frac{1}{n}(2\sin^2x)^n$

且 $0\le 2\sin^2x<1$. 设 $x_n=\frac{1}{n}(2\sin^2x)^n$.

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{(2\sin^2x)^{n+1}}{n+1}\cdot\frac{n}{(2\sin^2x)^n}=\lim_{n\to\infty}\frac{n}{n+1}\cdot 2\sin^2x$$

$$=2\sin^2x<1. \quad \therefore \sum_{n=1}^{\infty}\frac{1}{n}(2\sin^2x)^n \text{ 收敛.}$$

$\therefore$ 原级数绝对收敛.

(3) $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{n}{3^{n-1}}$

解：设 $x_n=(-1)^{n-1}\frac{n}{3^{n-1}}$. $\lim_{n\to\infty}\left|\frac{x_{n+1}}{x_n}\right| = \lim_{n\to\infty}\frac{n+1}{n}\cdot\frac{3^{n-1}}{3^n}=\frac{1}{3}<1$.

当 $x=k\pi\pm\frac{\pi}{4}$ 时，$\sin^2x=\frac{1}{2}$.

$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{2^n\sin^{2n}x}{n}=\sum_{n=1}^{\infty}\frac{(-1)^n}{n}$ $\therefore$ 条件收敛.

其它情况下 $\left|(-1)^{n+1}\frac{2^n\sin^{2n}x}{n}\right|=\frac{1}{n}(2\sin^2x)^2$. $2\sin^2x>1$.

级数的通项趋于无穷大. $\therefore$ 原级数发散.

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$\sum_{n=1}^{\infty}\left|(-1)^{n-1}\frac{n}{3^{n-1}}\right|$ 收敛. ∴ 原级数绝对收敛.

(4) $1-\frac{1}{2!}+\frac{1}{3}-\frac{1}{4!}+\frac{1}{5}-\cdots$

解: 设 $1-\frac{1}{2!}+\frac{1}{3}-\frac{1}{4!}+\frac{1}{5}-\cdots$ 部分和数列为 $\{S_n\}$, 则:

$S_{2n}=\sum_{k=1}^{n}\frac{1}{2k-1}-\sum_{k=1}^{n}\frac{1}{(2k)!}$ ∵ $\sum_{k=1}^{\infty}\frac{1}{(2n)!}$ 收敛, $\sum_{n=1}^{\infty}\frac{1}{(2n-1)}$ 发散.

∴ $\lim_{n\to\infty}S_{2n}=+\infty$. ∴ 原级数发散.

(5) $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n+x}$ $(x>0)$.

解: $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n+x}$ 是交错级数, 且 $\left\{\frac{1}{n+x}\right\}$ 单调趋于0. ∴ $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n+x}$ 收敛. (Leibniz级数).

而 $\left|\frac{(-1)^{n+1}}{n+x}\right|=\frac{1}{n+x}$ $(x>0)$. $\sum_{n=1}^{\infty}\frac{1}{n+x}$ 发散.

∴ 原级数条件收敛.

(6) $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}\cos\frac{n\pi}{3}$.

解: 设 $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}\cos\frac{n\pi}{3}$ 的部分和数列为 $\{S_n\}$, 则:

$S_{6n}=\sum_{k=1}^{2n}\frac{(-1)^{k-1}}{2\sqrt{3k-2}}+\sum_{k=1}^{2n}\frac{(-1)^{k}}{2\sqrt{3k-1}}+\sum_{k=1}^{2n}\frac{(-1)^{k}}{\sqrt{3k}}$.

$n=3k$: $\frac{1}{\sqrt{n}}\cos\frac{n\pi}{3}=\frac{1}{\sqrt{3k}}\cos k\pi=\frac{(-1)^k}{\sqrt{3k}}$.

$n=3k-1$: $\frac{1}{\sqrt{n}}\cos\frac{n\pi}{3}=\frac{1}{\sqrt{3k-1}}\cdot\cos\frac{(3k-1)\pi}{3}=\frac{1}{\sqrt{3k-1}}\left[\cos\left(k\pi-\frac{\pi}{3}\right)\right]=\frac{(-1)^k}{2\sqrt{3k-1}}$.

$n=3k-2$: $\frac{1}{\sqrt{n}}\cos\frac{n\pi}{3}=\frac{1}{\sqrt{3k-2}}\cdot\cos\frac{(3k-2)\pi}{3}=\frac{1}{\sqrt{3k-2}}\left[\cos\left(k\pi-\frac{2\pi}{3}\right)\right]=\frac{(-1)^{k-1}}{2\sqrt{3k-2}}$.

由于上述三个级数均为 Leibniz 级数, ∴ 都收敛. 即 $\lim_{n\to\infty}S_{6n}$ 存在且有限.

同理可证: $\lim_{n\to\infty}S_{6n+1}$, $\lim_{n\to\infty}S_{6n+2}$, $\lim_{n\to\infty}S_{6n+3}$, $\lim_{n\to\infty}S_{6n+4}$, $\lim_{n\to\infty}S_{6n+5}$ 存在且有限.

∴ 原级数收敛.

$\left|\frac{1}{\sqrt{n}}\cos\frac{n\pi}{3}\right| \geq \frac{1}{2\sqrt{n}}$. 而 $\sum_{n=1}^{\infty}\frac{1}{2\sqrt{n}}$ 发散. ∴ $\sum_{n=1}^{\infty}\left|\frac{1}{\sqrt{n}}\cos\frac{n\pi}{3}\right|$ 发散.

∴ $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}\cos\frac{n\pi}{3}$ 条件收敛.

(7) $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{2^{n^2}}{n!}$.

解: $\left|\frac{x_{n+1}}{x_n}\right| = \frac{2^{(n+1)^2}}{(n+1)!}\cdot\frac{n!}{2^{n^2}} = \frac{2^{(n+1)^2-n^2}}{n+1} = \frac{2^{2n+1}}{n+1} \to +\infty$. ∴ 发散.

(8) $\sum_{n=2}^{\infty}(-1)^n\frac{\ln^2 n}{n}$.

解: 当 $n\geq 8$ 时, $\frac{\ln^2 n}{n}$ 单调减为趋于0. ∴ $\sum_{n=2}^{\infty}(-1)^n\cdot\frac{\ln^2 n}{n}$ 为 Leibniz 级数. ∴收敛.

而 $\frac{\ln^2 n}{n} > \frac{1}{n}$ ⇒ $\sum_{n=2}^{\infty}\frac{\ln^2 n}{n}$ 发散. ∴ 原级数条件收敛.

(9) $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n^2}{2^n}x^n$.

解: $\lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\sqrt[n]{\frac{n^2}{2^n}|x|^n} = \lim_{n\to\infty}\frac{|x|}{2}\cdot\sqrt[n]{n^2} = \frac{|x|}{2}$.

∴ 当 $|x|<2$ 时, 原级数绝对收敛.

当 $|x|>2$ 时, 原级数发散. (通项 $\not\to 0$)

当 $|x|=2$ 时, 原级数为 $\sum_{n=1}^{\infty}(-1)^n n^2$ 或 $\sum_{n=1}^{\infty}n^2$ 也发散. (通项 $\not\to 0$)

(10) $\sum_{n=2}^{\infty}\frac{x^n}{n^p\ln^q n}$

解: $\lim_{n\to\infty}\sqrt[n]{|u_n|} = \lim_{n\to\infty}\sqrt[n]{\left|\frac{x^n}{n^p\ln^q n}\right|} = |x|$.

∴ 当 $|x|<1$ 时, 原级数绝对收敛.

当 $|x|>1$ 时, 原级数发散.

当 $x=1$ 时, $\sum_{n=2}^{\infty}\frac{x^n}{n^p\ln^q n} = \sum_{n=2}^{\infty}\frac{1}{n^p\ln^q n}$ ⇒

| $p>1$ 或 $p=1$ 且 $q>1$ 绝对收敛.
| 其它情况发散.

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当 $x=-1$ 时, $\sum_{n=2}^{\infty}\frac{x^n}{n^p\ln^q n}=\sum_{n=2}^{\infty}\frac{(-1)^n}{n^p\ln^q n}\Rightarrow$

- $p>1$ 或 $p=1$ 且 $q>1$. 绝对收敛
- $p=1$ 且 $q\le 1$ 或 $0<p<1$ 或 $p=0$ 且 $q>0$ 条件收敛.
- 其它情况发散.

(III) $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\cdot\frac{a}{1+a^n}$ $(a>0)$.

解: 当 $a>1$ 时, $\lim_{n\to\infty}\sqrt[n]{|u_n|}=\frac{1}{a}<1$ ∴ 原级数绝对收敛,

当 $a=1$ 时, $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\cdot\frac{a}{1+a^n}=\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{2n}$. ∴ 条件收敛.

当 $0<a<1$ 时. ∵ $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}$ 收敛. $\left\{\frac{a}{1+a^n}\right\}$ 单调有界.

由 Abel 判别法可知 $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\cdot\frac{a}{1+a^n}$ 收敛

但 $|u_n|\sim\frac{a}{n}$ $(n\to\infty)$ 发散. ∴ 原级数条件收敛.

∴ 当 $a>1$. 绝对收敛.

当 $0<a\le 1$. 条件收敛.

(IV) $\sum_{n=1}^{\infty}\frac{\sin(n+\frac{1}{n})}{n}$

解: $\frac{\sin(n+\frac{1}{n})}{n}=\frac{\sin n\cdot\cos\frac{1}{n}}{n}+\frac{\cos n\cdot\sin\frac{1}{n}}{n}$

∵ $\left|\frac{\cos n\cdot\sin\frac{1}{n}}{n}\right|\le\left|\frac{\sin\frac{1}{n}}{n}\right|\le\frac{1}{n^2}$ $(n\to\infty)$ ∴ $\sum_{n=1}^{\infty}\frac{\cos n\cdot\sin\frac{1}{n}}{n}$ 绝对收敛.

对于 $\sum_{n=1}^{\infty}\frac{\sin n\cdot\cos\frac{1}{n}}{n}$. ∵ $\sum_{n=1}^{\infty}\sin n$ 有界. 而 $\left(\frac{\cos\frac{1}{x}}{x}\right)'=\frac{1}{x^2}\left(\frac{1}{x^2}\sin\frac{1}{x}-\cos\frac{1}{x}\right)<0$ $(x\ge 2)$.

(当 $x>2$ 时, $\frac{1}{x}<\frac{1}{2}<\frac{\pi}{4}\Rightarrow\sin\frac{1}{x}<\cos\frac{1}{x}\Rightarrow\frac{1}{x^2}\sin\frac{1}{x}<\cos\frac{1}{x}$)

即 $\left\{\frac{\cos\frac{1}{n}}{n}\right\}$ 单调下降趋于0. ∴ $\sum_{n=1}^{\infty}\frac{\sin n\cdot\cos\frac{1}{n}}{n}$ 也收敛.

∴ 原级数 $\sum_{n=1}^{\infty}\frac{\sin(n+\frac{1}{n})}{n}$ 收敛.

11. 设 $x_n>0$, $\lim_{n\to\infty}x_n=0$. 问交错级数 $\sum_{n=1}^{\infty}(-1)^{n+1}x_n$ 是否收敛?

解: 不一定.

例如: $x_n = \begin{cases} \frac{1}{k}, & n=2k \\ \frac{1}{k^2}, & n=2k-1 \end{cases}$ 则 $x_n>0$ 且 $\lim_{n\to\infty} x_n = 0$.

但是级 $\sum_{n=1}^{\infty} (-1)^{n+1} x_n$ 发散.

12. 若级数 $\sum_{n=1}^{\infty} x_n$ 收敛, $\lim_{n\to\infty} \frac{x_n}{y_n} = 1$. 问级数 $\sum_{n=1}^{\infty} y_n$ 是否收敛? 举例说明:

$$x_n = \frac{(-1)^{n-1}}{\sqrt{n}} \qquad y_n = x_n + \frac{1}{n}.$$

解: 必一定收敛. 如上述所示 x_n, y_n. 是级 $\sum_{n=1}^{\infty} x_n$ 收敛 (Leibniz级数).

$$\lim_{n\to\infty} \frac{x_n}{y_n} = \lim_{n\to\infty} \frac{(-1)^{n-1}}{\sqrt{n}} \cdot \frac{n}{1+n\cdot\frac{(-1)^{n-1}}{\sqrt{n}}} = \lim_{n\to\infty} \frac{n\cdot(-1)^{n-1}}{\sqrt{n}+n\cdot(-1)^{n-1}}$$

$$= \lim_{n\to\infty} \frac{(-1)^{n-1}}{\frac{1}{\sqrt{n}}+(-1)^{n-1}} = 1.$$

但由于 $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散. $\therefore \sum_{n=1}^{\infty} y_n$ 发散. (原因: 非正项级数)

13. 设正项数列 $\{a_n\}$ 单调减少, 且 $\sum_{n=1}^{\infty} (-1)^n a_n$ 发散, 试问级数 $\sum_{n=1}^{\infty} \left(\frac{1}{1+a_n}\right)^n$ 是否收敛? 并说明理由.

解: $\sum_{n=1}^{\infty} \left(\frac{1}{1+a_n}\right)^n$ 收敛.

正项数列 $\{a_n\}$ 单调减少, 若有 $\lim_{n\to\infty} a_n = 0$. 则 $\sum_{n=1}^{\infty} (-1)^n a_n$ 是 Leibniz 级数 必收敛, 这与条件矛盾. $\therefore$ 必有 $\lim_{n\to\infty} a_n = \alpha > 0$.

当 n 充分大时, $\left(\frac{1}{1+a_n}\right)^n < \left(\frac{1}{1+\frac{\alpha}{2}}\right)^n$. 而 $\lim_{n\to\infty} \sqrt[n]{\left(\frac{1}{1+\frac{\alpha}{2}}\right)^n} = \frac{1}{1+\frac{\alpha}{2}} < 1$

$\therefore \sum_{n=1}^{\infty} \left(\frac{1}{1+a_n}\right)^n$ 收敛.

14. 设 $f(x)$ 在 $[-1,1]$ 上具有二阶连续导数, 且 $\lim_{x\to 0} \frac{f(x)}{x} = 0$. 证明级数 $\sum_{n=1}^{\infty} f\left(\frac{1}{n}\right)$ 绝对收敛.

证明: $\because \lim_{x\to 0} \frac{f(x)}{x} = 0$. $\therefore f(0) = 0$ $f'(0) = 0$. 则有:

$$f\left(\frac{1}{n}\right) \sim \frac{1}{2} f''(0) \cdot \frac{1}{n^2} \quad (n\to\infty) \quad (f(x) \text{ 在 } 0 \text{ 点处二阶 Taylor 展开}).$$

$\therefore \sum_{n=1}^{\infty} f\left(\frac{1}{n}\right)$ 绝对收敛.

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15. 设 $a_n=\int_0^{\frac{\pi}{4}} \tan^n x\,dx$, $n=1,2,\cdots$.

(1) 求级数 $\sum_{n=1}^{\infty}\frac{a_n+a_{n+2}}{n}$ 的和;

(2) 对于 $\lambda>0$, 证明级数 $\sum_{n=1}^{\infty}\frac{a_n}{n^{\lambda}}$ 收敛.

解: (1) $a_n+a_{n+2}=\int_0^{\frac{\pi}{4}}\tan^n x\,dx+\int_0^{\frac{\pi}{4}}\tan^{n+2}x\,dx=\int_0^{\frac{\pi}{4}}\tan^n x\cdot\sec^2 x\,dx$

$=\int_0^{\frac{\pi}{4}}\tan^n x\,d\tan x=\frac{1}{n+1}$

$\therefore \sum_{n=1}^{\infty}\frac{a_n+a_{n+2}}{n}=\sum_{n=1}^{\infty}\frac{1}{n(n+1)}=1$.

(2) 由 $a_n>0$, $a_n+a_{n+2}=\frac{1}{n+1}$ 可知 $a_n<\frac{1}{n+1}<\frac{1}{n}$. 则 $\frac{a_n}{n^{\lambda}}<\frac{1}{n^{1+\lambda}}$

$\because \sum_{n=1}^{\infty}\frac{1}{n^{1+\lambda}}$ $(\lambda>0)$ 收敛. $\therefore \sum_{n=1}^{\infty}\frac{a_n}{n^{\lambda}}$ 收敛.

16. 设级数 $\sum_{n=1}^{\infty}\frac{x_n}{n^{a_0}}$ 收敛. 证明当 $a>a_0$ 时, 级数 $\sum_{n=1}^{\infty}\frac{x_n}{n^a}$ 也收敛.

证明: $\sum_{n=1}^{\infty}\frac{x_n}{n^a}=\sum_{n=1}^{\infty}\left(\frac{x_n}{n^{a_0}}\cdot\frac{1}{n^{a-a_0}}\right)$. $\because \sum_{n=1}^{\infty}\frac{x_n}{n^{a_0}}$ 收敛, $\left\{\frac{1}{n^{a-a_0}}\right\}$ 单调有界. $(a>a_0)$

由 Abel 判别法可知 $\sum_{n=1}^{\infty}\frac{x_n}{n^a}$ 收敛.

17. 利用级数的 Cauchy 乘积证明 $\sum_{n=0}^{\infty}\frac{1}{n!}\cdot\sum_{n=0}^{\infty}\frac{(-1)^n}{n!}=1$.

证明: 设 $\sum_{n=0}^{\infty}\frac{1}{n!}\cdot\sum_{n=0}^{\infty}\frac{(-1)^n}{n!}=\sum_{n=0}^{\infty}C_n$. 显然有 $C_0=1$.

当 $n\geq 1$ 时, $C_n=\sum_{i+j=n}\frac{(-1)^j}{i!\cdot j!}=\frac{1}{n!}\cdot\sum_{i+j=n}\frac{n!}{i!\cdot j!}(-1)^j=\frac{1}{n!}(1-1)^n=0$.

$\therefore \sum_{n=0}^{\infty}\frac{1}{n!}\cdot\sum_{n=0}^{\infty}\frac{(-1)^n}{n!}=1$.

$\sum_{n=0}^{\infty}\frac{x^n}{n!}\cdot\sum_{n=0}^{\infty}\frac{y^n}{n!}=\sum_{n=0}^{\infty}\frac{(x+y)^n}{n!}$ 柯西乘积. 取 $x=1$, $y=-1$. 即得结论.

(绝对收敛) (绝对收敛)

§2. 幂级数

习题 P.203.

1. 求下列函数项级数的收敛域：

(1) $\sum_{n=1}^{\infty} \frac{1}{(1-x)^n}$.

解：$\sum_{n=1}^{\infty} \frac{1}{(1-x)^n} = \sum_{n=1}^{\infty} \left(\frac{1}{1-x}\right)^n$. 当 $\left|\frac{1}{1-x}\right| < 1$ 即 $x > 2$ 或 $x < 0$ 时收敛.

当 $x = 0$ 时，$\sum_{n=1}^{\infty} \frac{1}{(1-x)^n} = \sum_{n=1}^{\infty} 1^n$. 发散

当 $x = 2$ 时，$\sum_{n=1}^{\infty} \frac{1}{(1-x)^n} = \sum_{n=1}^{\infty} (-1)^n$. 发散

∴ 收敛域为 $(-\infty, 0) \cup (2, +\infty)$.

(2) $\sum_{n=1}^{\infty} 2^{nx}$

解：$\sum_{n=1}^{\infty} 2^{nx} = \sum_{n=1}^{\infty} (2^x)^n$. 当 $|2^x| < 1$ 即 $x < 0$ 时收敛.

当 $x = 0$ 时，$\sum_{n=1}^{\infty} 2^{nx} = \sum_{n=1}^{\infty} 1^n$. 发散.

∴ 收敛域为 $(-\infty, 0)$.

(3) $\sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\frac{x}{2}\right)^{2n}$.

解：级数为交错级数. 设 $u_n = \frac{1}{(n!)^2} \left(\frac{x}{2}\right)^{2n} \xrightarrow{单调} 0$. $(n \to \infty)$ $\forall x \in (-\infty, +\infty)$.

即原级数为 Leibniz 级数. 收敛. ∴ 收敛域为 $(-\infty, +\infty)$

2. 求下列幂级数的收敛半径与收敛域：

(1) $\sum_{n=1}^{\infty} \frac{(x-5)^n}{\sqrt{n}}$

解：$a_n = \frac{1}{\sqrt{n}}$. $A = \lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{1}{\sqrt[2n]{n}} = 1$. ∴ $R = 1$.

$|x-5| < 1 \Rightarrow 4 < x < 6$ 内收敛

当 $x = 4$ 时，$\sum_{n=1}^{\infty} \frac{(x-5)^n}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ 为 Leibniz 级数. 收敛.

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当 $x=6$ 时, $\sum_{n=1}^{\infty} \frac{(x-5)^n}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ 发散.

∴ 收敛半径 $R=1$, 收敛域 $[4,6)$.

(2) $\sum_{n=1}^{\infty} \frac{3^n}{n!}\left(\frac{x-1}{2}\right)^n$

解: $a_n = \frac{\left(\frac{3}{2}\right)^n}{n!}$ $A = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{\left(\frac{3}{2}\right)^{n+1}}{(n+1)!} \cdot \frac{n!}{\left(\frac{3}{2}\right)^n} = \frac{3}{2} \lim_{n\to\infty} \frac{1}{n+1} = 0$.

∴ $R=+\infty$. 收敛半径 $R=+\infty$, 收敛域 $(-\infty, +\infty)$.

(3) $\sum_{n=1}^{\infty} \frac{3^n+(-2)^n}{n} x^n$.

解: $a_n = \frac{3^n+(-2)^n}{n}$ $A = \overline{\lim_{n\to\infty}} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{3\cdot\sqrt[n]{1+\left(-\frac{2}{3}\right)^n}}{\sqrt[n]{n}} = 3$. $R=\frac{1}{3}$.

当 $x=-\frac{1}{3}$ 处. $\sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty} \frac{1}{n}\left[(-1)^n + \left(\frac{2}{3}\right)^n\right]$. 收敛. (两个级数分别收敛).

当 $x=\frac{1}{3}$ 处. $\sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty} \frac{1}{n}\left[1 + \left(-\frac{2}{3}\right)^n\right]$. 发散. (一个收敛, 一个发散).

∴ 收敛半径 $R=\frac{1}{3}$. 收敛域 $\left[-\frac{1}{3}, \frac{1}{3}\right)$.

(4) $\sum_{n=1}^{\infty} (-1)^n \frac{\ln(n+1)}{n+1} (x+1)^n$

解: $a_n = (-1)^n \frac{\ln(n+1)}{n+1}$ $A = \lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{\frac{\ln(n+1)}{n+1}} = 1$. (利用夹逼准则求极限)

∴ $R=\frac{1}{A}=1$. $|x+1|<1 \Rightarrow -2<x<0$ 收敛.

当 $x=0$ 时. $\sum_{n=1}^{\infty} a_n (x+1)^n = \sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (-1)^n \frac{\ln(n+1)}{n+1}$ Leibniz 收敛. 收敛.

当 $x=-2$ 时. $\sum_{n=1}^{\infty} a_n (x+1)^n = \sum_{n=1}^{\infty} (-1)^n a_n = \sum_{n=1}^{\infty} \frac{\ln(n+1)}{n+1} \left(> \sum_{n=1}^{\infty} \frac{1}{n+1}\right)$. 发散.

∴ 收敛半径 $R=1$. 收敛域 $(-2, 0]$.

(5) $\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^{-n^2} x^n$

解: $a_n = (1+\frac{1}{n})^{-n^2}$. $A = \lim\limits_{n\to\infty} \sqrt[n]{|a_n|} = \lim\limits_{n\to\infty} (1+\frac{1}{n})^{-n^2\cdot\frac{1}{n}} = \lim\limits_{n\to\infty} (1+\frac{1}{n})^{-n} = \frac{1}{e}$.

$\therefore R = \frac{1}{A} = e$. 在 $x=e$ 处: $\sum\limits_{n=1}^{\infty} a_n x^n = \sum\limits_{n=1}^{\infty} (1+\frac{1}{n})^{-n^2} e^n$. 通项不趋于0. 发散.

在 $x=-e$ 处: $\sum\limits_{n=1}^{\infty} a_n x^n = \sum\limits_{n=1}^{\infty} (1+\frac{1}{n})^{-n^2} (-e)^n$. 通项不趋于0. 发散.

$\therefore$ 收敛半径 $R=e$. 收敛域 $(-e, e)$.

(6) $\sum\limits_{n=1}^{\infty} (-1)^n \cdot \frac{x^{2n}}{n\cdot 2^n}$

解: $\sum\limits_{n=1}^{\infty} (-1)^n \cdot \frac{x^{2n}}{n\cdot 2^n} = \sum\limits_{n=1}^{\infty} a_n x^n$. ($a_{2k+1}=0$. 奇次项系数为0).

$A = \overline{\lim\limits_{n\to\infty}} \sqrt[n]{|a_n|} = \overline{\lim\limits_{n\to\infty}} \sqrt[2n]{\frac{1}{n\cdot 2^n}} = \frac{1}{\sqrt{2}}$.

$\therefore R = \frac{1}{A} = \sqrt{2}$. 在 $x = \pm\sqrt{2}$ 处. $\sum\limits_{n=1}^{\infty} a_n x^n = \sum\limits_{n=1}^{\infty} (-1)^n \cdot \frac{2^n}{n\cdot 2^n} = \sum\limits_{n=1}^{\infty} \frac{(-1)^n}{n}$ 收敛.

$\therefore$ 收敛半径 $R=\sqrt{2}$. 收敛域 $[-\sqrt{2}, \sqrt{2}]$.

(7) $\sum\limits_{n=1}^{\infty} (1+\frac{1}{2}+\cdots+\frac{1}{n})(x-1)^n$

解: $a_n = 1+\frac{1}{2}+\cdots+\frac{1}{n}$. $A = \lim\limits_{n\to\infty} \sqrt[n]{|a_n|} = \lim\limits_{n\to\infty} \sqrt[n]{1+\frac{1}{2}+\cdots+\frac{1}{n}} = 1$.

(夹逼性: $1 < 1+\frac{1}{2}+\cdots+\frac{1}{n} < n$).

$\therefore R = \frac{1}{A} = 1$. $|x-1|<1 \Rightarrow 0<x<2$ 收敛.

在 $x=0$ 处. $\sum\limits_{n=1}^{\infty} a_n (x-1)^n = \sum\limits_{n=1}^{\infty} (-1)^n (1+\frac{1}{2}+\cdots+\frac{1}{n})$. 通项不趋于0. 发散.

在 $x=2$ 处. $\sum\limits_{n=1}^{\infty} a_n (x-1)^n = \sum\limits_{n=1}^{\infty} (1+\frac{1}{2}+\cdots+\frac{1}{n})$. 通项不趋于0. 发散.

$\therefore$ 收敛半径 $R=1$. 收敛域 $(0, 2)$.

(8) $\sum\limits_{n=1}^{\infty} \frac{n!}{n^n} x^n$.

解: $a_n = \frac{n!}{n^n}$. $A = \lim\limits_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim\limits_{n\to\infty} \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \lim\limits_{n\to\infty} (\frac{n}{n+1})^n = \frac{1}{e}$.

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$\therefore R=\frac{1}{A}=e$. 当 $x=\pm e$ 时, $\sum_{n=1}^{\infty} a_n x^n=\sum_{n=1}^{\infty}\frac{n!}{n^n}(\pm e)^n$. 通项不趋于0. 发散.

$\therefore$ 收敛半径 $R=e$. 收敛域 $(-e, e)$.

(Stirling 公式: $n! \sim \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n}$ $n\to\infty$)

(9) $\sum_{n=1}^{\infty}\left(\frac{a^n}{n}+\frac{b^n}{n^2}\right)x^n$. $(a>b>0)$.

解: $a_n=\frac{a^n}{n}+\frac{b^n}{n^2}$. $A=\lim_{n\to\infty}\sqrt[n]{|a_n|}=\lim_{n\to\infty}\sqrt[n]{\left|\frac{a^n}{n}+\frac{b^n}{n^2}\right|}$

$=\lim_{n\to\infty}\frac{a}{\sqrt[n]{n}}\sqrt[n]{1+\frac{1}{n}\cdot\left(\frac{b}{a}\right)^n}=a$.

$\therefore R=\frac{1}{A}=\frac{1}{a}$. 当 $x=\frac{1}{a}$ 时, $\sum_{n=1}^{\infty}a_n x^n=\sum_{n=1}^{\infty}\left[\frac{1}{n}+\frac{1}{n^2}\left(\frac{b}{a}\right)^n\right]$. 发散 (一个发散, 一个收敛).

当 $x=-\frac{1}{a}$ 时. $\sum_{n=1}^{\infty}a_n x^n=\sum_{n=1}^{\infty}(-1)^n\left[\frac{1}{n}+\frac{1}{n^2}\left(\frac{b}{a}\right)^n\right]$. 收敛. (两个都收敛).

$\therefore$ 收敛半径 $R=\frac{1}{a}$. 收敛域 $\left[-\frac{1}{a}, \frac{1}{a}\right)$.

(10) $\sum_{n=1}^{\infty}\frac{x^n}{a^n+b^n}$. $(a>b>0)$.

解: $a_n=\frac{1}{a^n+b^n}$. $A=\lim_{n\to\infty}\sqrt[n]{|a_n|}=\lim_{n\to\infty}\sqrt[n]{\frac{1}{a^n+b^n}}=\frac{1}{a}\lim_{n\to\infty}\sqrt[n]{\frac{1}{1+\left(\frac{b}{a}\right)^n}}=\frac{1}{a}$.

$\therefore R=\frac{1}{A}=a$. 当 $x=\pm a$ 时, $\sum_{n=1}^{\infty}a_n x^n=\sum_{n=1}^{\infty}\frac{(\pm 1)^n}{a^n+b^n}$. 通项不趋于0. 发散.

$\therefore$ 收敛半径 $R=a$. 收敛域 $(-a, a)$.

3. 应用逐项求导或逐项积分求下列幂级数的和函数, 并指出它们的定义域:

(11) $\sum_{n=0}^{\infty}\frac{x^{2n}}{2n+1}$.

解: $\sum_{n=0}^{\infty}\frac{x^{2n}}{2n+1}=\sum_{n=0}^{\infty}a_n x^n$. ($a_{2k+1}=0$. 奇次项系数为0).

$A=\lim_{n\to\infty}\sqrt[n]{|a_n|}=\lim_{n\to\infty}\sqrt[2n]{\frac{1}{2n+1}}=1$. $\therefore R=\frac{1}{A}=1$.

在 $x=\pm 1$ 处，$\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} \frac{1}{2n+1}$ 发散，∴ 收敛域 $(-1,1)$.

设 $S(x)=\sum_{n=0}^{\infty} \frac{x^{2n}}{2n+1}$，则 $xS(x)=\sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1}$.

逐项求导得：$(xS(x))' = \sum_{n=0}^{\infty} x^{2n} = \frac{1}{1-x^2}$.

$$\therefore S(x)=\frac{1}{x}\int_0^x \frac{1}{1-t^2}dt = \frac{1}{2x}\int_0^x \left(\frac{1}{1+t}+\frac{1}{1-t}\right)dt$$

$$= \frac{1}{2x}\ln\frac{1+x}{1-x}, \quad x\in(-1,1).$$

(2) $\sum_{n=1}^{\infty}(n-1)x^n$.

解：$a_n = n-1$. $A=\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{n}{n-1}=1$. $\therefore R=\frac{1}{A}=1$.

在 $x=\pm 1$ 处，$\sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty}(n-1)\cdot(\pm 1)^n$. 通项不趋于0，发散，∴ 收敛域 $(-1,1)$.

设 $S(x)=\sum_{n=1}^{\infty}(n-1)x^n = \sum_{n=0}^{\infty} n x^{n+1} = x^2\sum_{n=1}^{\infty} n x^{n-1}$. $f(x)=\frac{S(x)}{x^2}=\sum_{n=1}^{\infty} n x^{n-1}$.

逐项积分：$\int_0^x f(t)dt = \sum_{n=1}^{\infty}\int_0^x n t^{n-1}dt = \sum_{n=1}^{\infty} x^n = \frac{1}{1-x}-1 = \frac{x}{1-x}$.

再逐项求导：$f(x)=\frac{S(x)}{x^2}=\left(\frac{x}{1-x}\right)' = \frac{1}{(1-x)^2}$.

$$\therefore S(x)=\frac{x^2}{(1-x)^2}, \quad x\in(-1,1).$$

(3) $\sum_{n=0}^{\infty}(2^{n+1}-1)x^n$.

解：$\sum_{n=0}^{\infty}(2^{n+1}-1)x^n = \sum_{n=0}^{\infty} 2\cdot(2x)^n - \sum_{n=0}^{\infty} x^n = 2\cdot\frac{1}{1-2x}-\frac{1}{1-x} = \frac{1}{(1-2x)(1-x)}$.

当 $\begin{cases}|2x|<1\\ |x|<1\end{cases}$ 即 $-\frac{1}{2}<x<\frac{1}{2}$ 时，$S(x)=\frac{1}{(1-2x)(1-x)}$.

$$\therefore S(x)=\frac{1}{(1-2x)(1-x)}, \quad x\in\left(-\frac{1}{2},\frac{1}{2}\right).$$

(4) $\sum_{n=1}^{\infty}(-1)^{n-1}n^2x^n$.

解: $a_n=(-1)^{n-1}n^2$. $A=\lim\limits_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim\limits_{n\to\infty}\frac{(n+1)^2}{n^2}=1$. $\therefore R=\frac{1}{A}=1$.

当 $x=\pm1$ 处, $\sum\limits_{n=1}^{\infty}a_nx^n=\sum\limits_{n=1}^{\infty}(-1)^{n-1}n^2\cdot(\pm1)^n$. 通项不趋于0, 发散. $\therefore$ 收敛域 $(-1,1)$.

设 $S(x)=\sum\limits_{n=1}^{\infty}(-1)^{n-1}\cdot n^2x^n$. $f(x)=\frac{S(x)}{x}=\sum\limits_{n=1}^{\infty}(-1)^{n-1}n^2x^{n-1}=\sum\limits_{n=1}^{\infty}n^2(-x)^{n-1}$.

两边积分: $\int_0^x f(t)dt=\sum\limits_{n=1}^{\infty}\int_0^x n^2(-t)^{n-1}dt=-\sum\limits_{n=1}^{\infty}n(-x)^n=x\sum\limits_{n=1}^{\infty}n(-x)^{n-1}$.

设 $g(x)=\int_0^x f(t)dt=x\sum\limits_{n=1}^{\infty}n(-x)^{n-1}$. $\frac{g(x)}{x}=\sum\limits_{n=1}^{\infty}n(-x)^{n-1}$.

两边积分: $\int_0^x\frac{g(t)}{t}dt=\sum\limits_{n=1}^{\infty}\int_0^x n(-t)^{n-1}dt=-\sum\limits_{n=1}^{\infty}(-x)^n=1-\frac{1}{1+x}$.

两边求导: $\frac{g(x)}{x}=\frac{1}{(1+x)^2}\Rightarrow g(x)=\frac{x}{(1+x)^2}$.

$f(x)=g'(x)=\left[\frac{x}{(1+x)^2}\right]'=\frac{1-x}{(1+x)^3}=\frac{S(x)}{x}$.

$\therefore S(x)=\frac{x(1-x)}{(1+x)^3}$. $x\in(-1,1)$.

(5) $\sum\limits_{n=1}^{\infty}n(n+1)x^n$.

解: $a_n=n(n+1)$. $A=\lim\limits_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim\limits_{n\to\infty}\frac{(n+1)(n+2)}{n(n+1)}=\lim\limits_{n\to\infty}\frac{n+2}{n}=1$. $\therefore R=1$.

当 $x=\pm1$ 处. $\sum\limits_{n=1}^{\infty}a_nx^n=\sum\limits_{n=1}^{\infty}n(n+1)\cdot(\pm1)^n$. 通项不趋于0. 发散. $\therefore$ 收敛域 $(-1,1)$.

设 $S(x)=\sum\limits_{n=1}^{\infty}n(n+1)x^n$. $\frac{S(x)}{x}=\sum\limits_{n=1}^{\infty}n(n+1)x^{n-1}$.

两边积分: $\int_0^x\frac{S(t)}{t}dt=\sum\limits_{n=1}^{\infty}(n+1)x^n=f(x)$.

再积分一次: $\int_0^x f(t)dt=\sum\limits_{n=1}^{\infty}x^{n+1}=\frac{1}{1-x}-1-x$.

两边求导: $f(x)=\frac{1}{(1-x)^2}-1$.

再求导一次: $\frac{S(x)}{x}=f'(x)=\frac{2}{(1-x)^3}$.

$\therefore S(x)=\frac{2x}{(1-x)^3}$. $x\in(-1,1)$.

(b) $\sum_{n=2}^{\infty} (-1)^n \cdot \frac{1}{n(n-1)} x^n$.

解: $a_n = (-1)^n \cdot \frac{1}{n(n-1)}$. $A = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{n(n-1)}{(n+1)\cdot n} = \lim_{n\to\infty} \frac{n-1}{n+1} = 1$. $\therefore R = 1$.

当 $x = \pm 1$ 处, $\sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty} (-1)^n \cdot \frac{1}{n(n-1)} (\pm 1)^n$ 收敛. $\therefore$ 收敛域 $[-1, 1]$.

设 $S(x) = \sum_{n=2}^{\infty} (-1)^n \cdot \frac{1}{n(n-1)} x^n = \sum_{n=2}^{\infty} \frac{(-x)^n}{n-1} - \sum_{n=2}^{\infty} \frac{(-x)^n}{n}$

$= -x \sum_{n=2}^{\infty} \frac{(-x)^{n-1}}{n-1} - \sum_{n=2}^{\infty} \frac{(-x)^n}{n} = -x f(x) - g(x)$.

$f(x) = \sum_{n=2}^{\infty} \frac{(-x)^{n-1}}{n-1}$. $f'(x) = -\sum_{n=2}^{\infty} (-x)^{n-2} = -\sum_{n=0}^{\infty} (-x)^n = -\frac{1}{1+x}$.

$g(x) = \sum_{n=2}^{\infty} \frac{(-x)^n}{n}$. $g'(x) = -\sum_{n=2}^{\infty} (-x)^{n-1} = -\sum_{n=1}^{\infty} (-x)^{n-1} = -1 + \frac{1}{1+x}$.

由于 $f(0) = g(0) = 0$. $\therefore f(x) = -\ln(1+x)$. $g(x) = -\ln(1+x) + x$.

$\therefore S(x) = x\ln(1+x) + \ln(1+x) - x$

$= (1+x)\ln(1+x) - x$. $\quad x \in [-1, 1]$.

(7) $\sum_{n=1}^{\infty} \frac{n}{n+1} x^n$.

解: $a_n = \frac{n}{n+1}$. $A = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{n+1}{n+2} \cdot \frac{n+1}{n}\right| = \lim_{n\to\infty} \frac{1 - \frac{1}{n+2}}{1 - \frac{1}{n+1}} = 1$. $\therefore R = 1$.

当 $x = \pm 1$ 处, $\sum_{n=1}^{\infty} a_n x_n = \sum_{n=1}^{\infty} \frac{n}{n+1} (\pm 1)^n$. 通项不趋于 0, 发散. $\therefore$ 收敛域 $(-1, 1)$.

设 $S(x) = \sum_{n=1}^{\infty} \frac{n}{n+1} x^n = \sum_{n=1}^{\infty} \left(1 - \frac{1}{n+1}\right) x^n = \sum_{n=1}^{\infty} x^n - \sum_{n=1}^{\infty} \frac{x^n}{n+1}$.

$= \left(\sum_{n=0}^{\infty} x^n - 1\right) - \frac{1}{x} \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1} = \frac{x}{1-x} - \frac{1}{x} \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1}$.

记 $f(x) = x\left[\frac{x}{1-x} - S(x)\right] = \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1}$.

两边求导: $f'(x) = \sum_{n=1}^{\infty} x^n = \frac{1}{1-x} - 1$.

两边积分: 由 $f(0) = 0 \Rightarrow f(x) = -\ln(1-x) - x = x\left[\frac{x}{1-x} - S(x)\right]$.

当 $x \neq 0$ 时, $S(x) = \frac{x}{1-x} + \frac{1}{x}\ln(1-x) + 1 = \frac{1}{1-x} + \frac{1}{x}\ln(1-x)$. $x \in (-1, 0) \cup (0, 1)$.

当 $x = 0$ 时, $S(x) = 0$.

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(8) $\sum_{n=1}^{\infty} \frac{2n-1}{2^n} x^{2n-2}$.

解: $a_n = \frac{2n-1}{2^n}$. $\sum_{n=1}^{\infty} \frac{2n-1}{2^n} x^{2n-2} = \sum_{n=1}^{\infty} a_n (x^2)^{n-1} = \frac{1}{x^2} \sum_{n=1}^{\infty} a_n (x^2)^n$.

$A = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{2n+1}{2^{n+1}} \cdot \frac{2^n}{2n-1} = \lim_{n\to\infty} \frac{1}{2} \cdot \frac{2n+1}{2n-1} = \frac{1}{2}$. $\therefore R = \frac{1}{A} = 2$.

$|x^2| < 2 \Rightarrow -\sqrt{2} < x < \sqrt{2}$. 收敛.

在 $x = \pm\sqrt{2}$ 处. $\sum_{n=1}^{\infty} a_n x^{2n-2} = \sum_{n=1}^{\infty} \frac{2n-1}{2^n} \cdot 2^{n-1} = \frac{1}{2} \sum_{n=1}^{\infty} (2n-1)$.

通项不趋于0. 发散. $\therefore$ 收敛域 $(-\sqrt{2}, \sqrt{2})$.

设 $S(x) = \sum_{n=1}^{\infty} \frac{2n-1}{2^n} x^{2n-2} = \frac{1}{x^2} \sum_{n=1}^{\infty} (2n-1) \cdot \left(\frac{x^2}{2}\right)^n = \frac{2}{x^2} \sum_{n=1}^{\infty} n \cdot \left(\frac{x^2}{2}\right)^n - \frac{1}{x^2} \sum_{n=1}^{\infty} \left(\frac{x^2}{2}\right)^n$

$= \sum_{n=1}^{\infty} n \left(\frac{x^2}{2}\right)^{n-1} - \frac{1}{x^2} \left(\frac{1}{1-\frac{x^2}{2}} - 1 \right) = \sum_{n=1}^{\infty} n \left(\frac{x^2}{2}\right)^{n-1} - \frac{1}{2-x^2}$.

$f(x) = S(x) + \frac{1}{2-x^2} = \sum_{n=1}^{\infty} n \left(\frac{x^2}{2}\right)^{n-1} = \sum_{n=1}^{\infty} n y^{n-1} = g(y)$. $\left(y = \frac{x^2}{2}\right)$.

两边积分: $\int_0^y g(t)\,dt = \sum_{n=1}^{\infty} y^n = \frac{1}{1-y} - 1$.

两边求导: $g(y) = \frac{1}{(1-y)^2}$. 即 $f(x) = g(y) = g\left(\frac{x^2}{2}\right) = \frac{1}{\left(1-\frac{x^2}{2}\right)^2} = \frac{4}{(2-x^2)^2}$.

$\therefore S(x) = f(x) - \frac{1}{2-x^2} = \frac{4}{(2-x^2)^2} - \frac{1}{2-x^2} = \frac{2+x^2}{(2-x^2)^2}$. $x \in (-\sqrt{2}, 0) \cup (0, \sqrt{2})$.

当 $x = 0$ 时. $S(x) = 0$.

4. 证明:

(1) $y = \sum_{n=0}^{\infty} \frac{x^{4n}}{(4n)!}$ 满足方程 $y^{(4)} = y$;

证明: $y^{(4)} = \sum_{n=1}^{\infty} \frac{4n \times (4n-1) \times (4n-2) \times (4n-3)}{(4n)!} x^{(4n-4)} = \sum_{n=1}^{\infty} \frac{x^{4(n-1)}}{4(n-1)!}$

$= \sum_{n=0}^{\infty} \frac{x^{4n}}{4n!} = y$.

(2) $y = \sum_{n=0}^{\infty} \frac{x^n}{(n!)^2}$ 满足方程 $xy'' + y' - y = 0$.

证明：$y' = \sum_{n=1}^{\infty} \frac{n x^{n-1}}{n! \cdot n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{n! \cdot (n-1)!} = \sum_{n=0}^{\infty} \frac{x^n}{(n+1)! \cdot n!}$.

$y'' = \sum_{n=1}^{\infty} \frac{n x^{n-1}}{(n+1)! \cdot (n!)^2} = \sum_{n=0}^{\infty} \frac{(n+1) x^n}{(n+2)! \cdot [(n+1)!]^2}$

$xy'' + y' - y = \sum_{n=1}^{\infty} \frac{n x^n}{(n+1)! \cdot (n!)^2} + \sum_{n=0}^{\infty} \frac{x^n}{(n+1)! \cdot (n!)^2} - \sum_{n=0}^{\infty} \frac{x^n}{(n!)^2}$

$= \sum_{n=1}^{\infty} \frac{n x^n}{(n+1)! \cdot (n!)^2} + \sum_{n=0}^{\infty} \frac{-n x^n}{(n+1)! \cdot (n!)^2}$

$= \sum_{n=1}^{\infty} \frac{n x^n}{(n+1)! \cdot (n!)^2} - \sum_{n=1}^{\infty} \frac{n x^n}{(n+1)! \cdot (n!)^2} = 0$.

5. 应用幂级数性质求下列级数之和：

(1) $\sum_{n=0}^{\infty} \frac{(n+1)^2}{2^n}$.

解：设 $f(x) = \sum_{n=0}^{\infty} (n+1)^2 x^n$.

逐项积分：$\int_0^x f(t)\,dt = \sum_{n=0}^{\infty} (n+1) \cdot x^{n+1} = \sum_{n=1}^{\infty} n x^n = x \sum_{n=1}^{\infty} n x^{n-1} = g(x)$.

$\frac{g(x)}{x} = \sum_{n=1}^{\infty} n x^{n-1}$.

逐项积分：$\int_0^x \frac{g(t)}{t}\,dt = \sum_{n=1}^{\infty} x^n = \frac{1}{1-x} - 1 \Rightarrow \frac{g(x)}{x} = \frac{1}{(1-x)^2} \Rightarrow g(x) = \frac{x}{(1-x)^2}$.

$f(x) = g'(x) = \frac{1+x}{(1-x)^3}$. $\therefore \sum_{n=0}^{\infty} \frac{(n+1)^2}{2^n} = f(\frac{1}{2}) = (1+\frac{1}{2}) \cdot 8 = 12$.

(2) $\sum_{n=0}^{\infty} (-1)^n \cdot \frac{1}{3^n (2n+1)}$.

解：设 $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \cdot x^{2n}$. $\Rightarrow x f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$.

逐项求导：$(x f(x))' = \sum_{n=0}^{\infty} (-1)^n \cdot x^{2n} = \sum_{n=0}^{\infty} (-x^2)^n = \frac{1}{1+x^2}$.

$\Rightarrow x f(x) = \arctan x$. $\Rightarrow f(x) = \frac{\arctan x}{x}$.

$\therefore \sum_{n=0}^{\infty} (-1)^n \cdot \frac{1}{3^n (2n+1)} = f(\frac{1}{\sqrt{3}}) = \frac{\arctan \frac{\sqrt{3}}{3}}{\frac{\sqrt{3}}{3}} = \frac{\pi}{6} \cdot \sqrt{3} = \frac{\sqrt{3}}{6}\pi$.

(3) $\sum_{n=1}^{\infty} \frac{1}{n\cdot 2^n}$.

解: 设 $f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n}$.

逐项求导: $f'(x) = \sum_{n=1}^{\infty} x^{n-1} = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$.

$$f(x) = \int_0^x \frac{1}{1-t}\,dt = -\ln|1-x| = \ln\frac{1}{1-x}, \qquad x \in [-1, 1),$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n\cdot 2^n} = f\left(\frac{1}{2}\right) = \ln\frac{1}{1-\frac{1}{2}} = \ln 2.$$

(4) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{3n-1}$.

解: 设 $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{3n-1} x^{3n} = x\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{3n-1} x^{3n-1}$

$$g(x) = \frac{f(x)}{x} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{3n-1} x^{3n-1}.$$

逐项求导: $g'(x) = \sum_{n=1}^{\infty} (-1)^{n+1} x^{3n-2} = -\frac{1}{x^2}\sum_{n=1}^{\infty} (-x^3)^n = -\frac{1}{x^2}\left(\frac{1}{1+x^3} - 1\right)$.

$$= -\frac{1}{x^2}\cdot\frac{-x^3}{1+x^3} = \frac{x}{1+x^3}.$$

$$g(x) = \int_0^x \frac{t}{1+t^3}\,dt = \frac{1}{6}\ln\left[\frac{4}{3}\left(x-\frac{1}{2}\right)^2 + 1\right] - \frac{1}{6}\ln\frac{4}{3} + \frac{1}{\sqrt{3}}\arctan\frac{2}{\sqrt{3}}\left(x-\frac{1}{2}\right)$$
$$+ \frac{\sqrt{3}}{18}\pi - \frac{1}{3}\ln(1+x).$$

$$\therefore \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{3n-1} = g(-1) = f(-1) = \frac{1}{6}\ln\frac{4}{3} - \frac{1}{6}\ln\frac{4}{3} + \frac{1}{\sqrt{3}}\arctan\frac{1}{\sqrt{3}} + \frac{\sqrt{3}}{18}\pi - \frac{1}{3}\ln 2$$
$$= \frac{\sqrt{3}}{9}\pi - \frac{1}{3}\ln 2.$$

(5) $\sum_{n=1}^{\infty} \frac{n(n+2)}{4^{n+1}}$.

解: 设 $f(x) = \sum_{n=1}^{\infty} n(n+2)x^{n+1}$.

逐项积分: $\int_0^x f(t)\,dt = \sum_{n=1}^{\infty} n x^{n+2} = x^3\sum_{n=1}^{\infty} n x^{n-1}$. 令 $g(x) = \sum_{n=1}^{\infty} n x^{n-1}$.

逐项积分: $\int_0^x g(t)\,dt = \sum_{n=1}^{\infty} x^n = \frac{1}{1-x} - 1 \Rightarrow g(x) = \sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2}$.

$$\Rightarrow \int_0^x f(t)\,dt = x^3 g(x) = \frac{x^3}{(1-x)^2} \Rightarrow f(x) = \frac{x^2(3-x)}{(1-x)^3}.$$

$$\therefore \sum_{n=1}^{\infty} \frac{n(n+2)}{4^{n+1}} = f\left(\frac{1}{4}\right) = \frac{\frac{1}{16}\cdot\frac{11}{4}}{\left(\frac{3}{4}\right)^3} = \frac{11}{27}.$$

(b) $\sum_{n=2}^{\infty} \frac{1}{(n^2-1)2^n}$.

解: 设 $f(x) = \sum_{n=2}^{\infty} \frac{x^n}{n^2-1} = \sum_{n=2}^{\infty} \frac{1}{2}\left(\frac{1}{n-1} - \frac{1}{n+1}\right)x^n = \frac{1}{2}\sum_{n=2}^{\infty}\frac{x^n}{n-1} - \frac{1}{2}\sum_{n=2}^{\infty}\frac{x^n}{n+1}$

$$= \frac{x}{2}\sum_{n=2}^{\infty}\frac{x^{n-1}}{n-1} - \frac{1}{2x}\sum_{n=2}^{\infty}\frac{x^{n+1}}{n+1} = \frac{x}{2}g(x) - \frac{1}{2x}h(x).$$

$$g(x) = \sum_{n=2}^{\infty}\frac{x^{n-1}}{n-1} \Rightarrow g'(x) = \sum_{n=2}^{\infty} x^{n-2} = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \Rightarrow g(x) = \ln\frac{1}{1-x}.$$

$$h(x) = \sum_{n=2}^{\infty}\frac{x^{n+1}}{n+1} \Rightarrow h'(x) = \sum_{n=2}^{\infty} x^n = \frac{1}{1-x} - 1 - x \Rightarrow h(x) = \ln\frac{1}{1-x} - x - \frac{1}{2}x^2.$$

$$\therefore f(x) = \frac{x}{2}g(x) - \frac{1}{2x}h(x) = \frac{x}{2}\ln\frac{1}{1-x} - \frac{1}{2x}\left(\ln\frac{1}{1-x} - x - \frac{1}{2}x^2\right)$$

$$= \frac{x^2-1}{2x}\ln\frac{1}{1-x} + \frac{1}{2} + \frac{x}{4} = \frac{1-x^2}{2x}\ln(1-x) + \frac{x}{4} + \frac{1}{2}.$$

$$\therefore \sum_{n=2}^{\infty}\frac{1}{(n^2-1)2^n} = f\left(\frac{1}{2}\right) = \frac{1-\frac{1}{4}}{2\cdot\frac{1}{2}}\ln\left(1-\frac{1}{2}\right) + \frac{1}{4}\cdot\frac{1}{2} + \frac{1}{2} = -\frac{3}{4}\ln 2 + \frac{5}{8}.$$

6. 求下列函数在指定点 x_0 处 Taylor 展开，并确定它的收敛范围：

(1) $1+2x-3x^2+5x^3$, $x_0=1$

解: 令 $x-1=t$, 则有 $x=t+1$

$$1+2x-3x^2+5x^3 = 1+2(t+1)-3(t+1)^2+5(t+1)^3 = 5+11t+12t^2+5t^3$$

$$= 5+11(x-1)+12(x-1)^2+5(x-1)^3. \qquad x\in(-\infty,+\infty)$$

(2) $\frac{x}{2-x-x^2}$, $x_0=0$.

解: $\frac{x}{2-x-x^2} = \frac{x}{(2+x)(1-x)} = \frac{1}{3}\left(\frac{1}{1-x} - \frac{2}{2+x}\right) = \frac{1}{3}\left(\sum_{n=0}^{\infty}x^n\right) - \frac{1}{3}\left[\sum_{n=0}^{\infty}\left(-\frac{x}{2}\right)^n\right]$

$$= \frac{1}{3}\sum_{n=0}^{\infty}\left[1 + \frac{(-1)^{n+1}}{2^n}\right]x^n. \qquad a_n = 1 + \frac{(-1)^{n+1}}{2^n}$$

$$A = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{1+\frac{(-1)^{n+2}}{2^{n+1}}}{1+\frac{(-1)^{n+1}}{2^n}} = \lim_{n\to\infty}\frac{1-\left(-\frac{1}{2}\right)^{n+1}}{1-\left(-\frac{1}{2}\right)^n} = 1.$$

~~在 $x=\pm 1$ 处: $\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty}\left(1+\frac{(-1)^{n+1}}{2^n}\right)\cdot(\pm 1)^n$ 通项不趋于 0, 发散.~~

$\therefore$ 收敛域 $(-1,1)$.

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(3) $(1+x)\ln(1-x)$. $x_0=0$.

解: $(1+x)\ln(1-x) = (1+x)\sum_{n=1}^{\infty}\left(-\frac{1}{n}x^n\right) = \sum_{n=1}^{\infty}\left(-\frac{x^n}{n}\right) - \sum_{n=1}^{\infty}\frac{x^{n+1}}{n}$

$= -x - \sum_{n=2}^{\infty}\frac{x^n}{n} - \sum_{n=2}^{\infty}\frac{x^n}{n-1} = -x - \sum_{n=2}^{\infty}\left(\frac{1}{n}+\frac{1}{n-1}\right)x^n$

$= -x - \sum_{n=2}^{\infty}\frac{2n-1}{n(n-1)}x^n$.

$a_n = \frac{2n-1}{n(n-1)}$. $A = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{2n+1}{(n+1)\cdot n}\cdot\frac{n(n-1)}{2n-1} = 1$. $\therefore R = \frac{1}{A} = 1$.

将 $x=1$ 处. $\sum_{n=2}^{\infty}\frac{2n-1}{n(n-1)} = \sum_{n=2}^{\infty}\left(\frac{1}{n}+\frac{1}{n-1}\right)$ 发散. (两个收敛级数的发散)

将 $x=-1$ 处. $\sum_{n=2}^{\infty}\frac{2n-1}{n(n-1)} = \sum_{n=2}^{\infty}\left[\frac{(-1)^n}{n} + \frac{(-1)^n}{n-1}\right]$ 收敛. (两个为 Leibniz 级数)

$\therefore (1+x)\ln(1-x) = -x - \sum_{n=2}^{\infty}\frac{2n-1}{n(n-1)}x^n$. 收敛域 $[-1,1)$.

(4) $\ln x$. $x_0=2$.

解: $\ln x = \ln[2+(x-2)] = \ln 2 + \ln\left(1+\frac{x-2}{2}\right) = \ln 2 + \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\cdot\left(\frac{x-2}{2}\right)^n$

$= \ln 2 + \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n\cdot 2^n}(x-2)^n$.

$a_n = \frac{(-1)^{n+1}}{n\cdot 2^n}$. $A = \lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\frac{1}{2}\cdot\frac{1}{\sqrt[n]{n}} = \frac{1}{2}$. $\therefore R = \frac{1}{A} = 2$.

$|x-2|<2 \Rightarrow 0<x<4$

将 $x=0$ 处. $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n\cdot 2^n}(-2)^n = \sum_{n=1}^{\infty}\left(-\frac{1}{n}\right)$. 发散.

将 $x=4$ 处. $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n\cdot 2^n}2^n = \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}$. 收敛 (Leibniz 级数)

$\therefore \ln x = \ln 2 + \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n\cdot 2^n}(x-2)^n$. 收敛域 $(0,4]$.

(5) $x\arctan x - \ln\sqrt{1+x^2}$. $x_0=0$.

解: $x\arctan x - \ln\sqrt{1+x^2} = x\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2n-1}x^{2n-1} - \frac{1}{2}\ln(1+x^2)$

$= \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2n-1}x^{2n} - \frac{1}{2}\cdot\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}x^{2n}$

$= \sum_{n=1}^{\infty}(-1)^{n-1}\cdot\left(\frac{1}{2n-1}-\frac{1}{2n}\right)x^{2n} = \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2n\cdot(2n-1)}x^{2n}$.

$a_n = \frac{(-1)^{n-1}}{2n\cdot(2n-1)}$. $A = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{2n\cdot(2n-1)}{2(n+1)\cdot(2n+1)} = \lim_{n\to\infty}\frac{4n^2-2n}{4n^2+6n+2} = 1$.

$\therefore R = \frac{1}{A} = 1$. 在 $x=\pm1$ 处, $\sum_{n=1}^{\infty} a_n x^{2n} = \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2n\cdot(2n-1)} = \sum_{n=1}^{\infty}\left(\frac{(-1)^{n-1}}{2n-1} - \frac{(-1)^{n-1}}{2n}\right)$

为 Leibniz 级数. 收敛.

$\therefore$ $x\arctan x - \ln\sqrt{1+x^2} = \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2n\cdot(2n-1)}x^{2n}$. 收敛域 $[-1,1]$.

(b) $\sqrt[3]{4-x^2}$. $x_0=0$.

解: $\sqrt[3]{4-x^2} = \sqrt[3]{4}\cdot\sqrt[3]{1-\left(\frac{x}{2}\right)^2} = \sqrt[3]{4}\cdot\sum_{n=0}^{\infty}\frac{(-1)^n}{2^{2n}}\binom{\frac{1}{3}}{n}x^{2n}$.

$a_n = \frac{(-1)^n}{2^{2n}}\binom{\frac{1}{3}}{n}$. $A = \lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\sqrt[2n]{\frac{1}{2^{2n}}\binom{\frac{1}{3}}{n}} = \frac{1}{2}$.

$\therefore R = \frac{1}{A} = 2$. 在 $x=\pm2$ 处, $\sum_{n=0}^{\infty} a_n\cdot x^{2n} = \sum_{n=0}^{\infty}\frac{(-1)^n}{2^{2n}}\binom{\frac{1}{3}}{n}\cdot 2^{2n} = \sum_{n=0}^{\infty}(-1)^n\binom{\frac{1}{3}}{n}$

收敛.

$\therefore$ $\sqrt[3]{4-x^2} = \sqrt[3]{4}\cdot\sum_{n=0}^{\infty}\frac{(-1)^n}{2^{2n}}\binom{\frac{1}{3}}{n}x^{2n}$. 收敛域 $[-2,2]$.

(7) $\sin x$, $x_0 = \frac{\pi}{6}$.

解: $\sin x = \sin\left[\left(x-\frac{\pi}{6}\right)+\frac{\pi}{6}\right] = \sin\frac{\pi}{6}\cdot\cos\left(x-\frac{\pi}{6}\right) + \cos\frac{\pi}{6}\cdot\sin\left(x-\frac{\pi}{6}\right)$.

$= \frac{1}{2}\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n)!}\left(x-\frac{\pi}{6}\right)^{2n} + \frac{\sqrt{3}}{2}\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}\left(x-\frac{\pi}{6}\right)^{2n+1}$. 收敛域 $(-\infty,+\infty)$.

(8) $\frac{x-1}{x+1}$. $x_0=1$.

解: $\frac{x-1}{x+1} = \frac{1}{2}\cdot\frac{x-1}{1+\frac{x-1}{2}} = \frac{x-1}{2}\cdot\sum_{n=0}^{\infty}\left(-\frac{x-1}{2}\right)^n = \frac{x-1}{2}\cdot\sum_{n=0}^{\infty}\frac{(-1)^n}{2^n}(x-1)^n$.

$= \sum_{n=0}^{\infty}\frac{(-1)^n}{2^{n+1}}\cdot(x-1)^{n+1} = \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2^n}(x-1)^n$.

$a_n = \frac{(-1)^{n-1}}{2^n}$ $A = \lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\frac{1}{2} = \frac{1}{2}$. $\therefore R = \frac{1}{A} = 2$. $|x-1|<2$ $\Rightarrow -1<x<3$.

在 $x=-1$ 处. $\sum_{n=1}^{\infty} a_n(x-1)^n = \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2^n}\cdot(-2)^n = \sum_{n=1}^{\infty}(-1)$. 发散.

在 $x=3$ 处. $\sum_{n=1}^{\infty} a_n(x-1)^n = \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2^n}\cdot 2^n = \sum_{n=1}^{\infty}(-1)^{n-1}$. 发散.

$\therefore$ $\frac{x-1}{x+1} = \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2^n}(x-1)^n$. 收敛域 $(-1,3)$.

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(9) $\int_0^x \frac{\sin t}{t}\,dt$. $x_0=0$.

解: $\int_0^x \frac{\sin t}{t}\,dt = \int_0^x \frac{1}{t}\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}t^{2n+1}\,dt = \int_0^x \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}t^{2n}\,dt$

$= \sum_{n=0}^{\infty}\int_0^x \frac{(-1)^n}{(2n+1)!}t^{2n}\,dt = \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}\cdot\frac{x^{2n+1}}{2n+1}$

$= \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2\cdot(2n)!}x^{2n+1}$. 收敛域 $(-\infty,+\infty)$.

(10) $\frac{e^{-x}}{1-x}$. $x_0=0$. (Cauchy 乘积)

解: $\frac{e^{-x}}{1-x} = \sum_{n=0}^{\infty}\frac{(-x)^n}{n!}\cdot\sum_{n=0}^{\infty}x^n = \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n}\frac{(-x)^k}{k!}\cdot x^{n-k}\right) = \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n}\frac{(-1)^k}{k!}\right)x^n$

$= 1+\sum_{k=0}^{1}\frac{(-1)^k}{k!}x + \sum_{n=2}^{\infty}\left(\sum_{k=0}^{n}\frac{(-1)^k}{k!}\right)x^n$

$= 1+(x-x)+\sum_{n=2}^{\infty}\left(1-1+\sum_{k=2}^{n}\right)x^n = 1+\sum_{n=2}^{\infty}\left(\sum_{k=2}^{n}\frac{(-1)^k}{k!}\right)x^n$.

$a_n = \sum_{k=2}^{n}\frac{(-1)^k}{k!}$. 易证当 $n\geq 4$ 时: $\frac{1}{2!}-\frac{1}{3!} < a_n < \frac{1}{2!}$

$\left(\frac{1}{2!}-\frac{1}{3!} < \frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}+\cdots\frac{(-1)^{n-1}}{(n-1)!}+\frac{(-1)^n}{n!} < \frac{1}{2!}\right)$

$\therefore A = \lim_{n\to\infty}\sqrt[n]{|a_n|} = 1$. $R=\frac{1}{A}=1$.

在 $x=\pm 1$ 处, $\sum_{n=2}^{\infty}a_n x^n = \sum_{n=2}^{\infty}\left(\sum_{k=2}^{n}\frac{(-1)^k}{k!}\right)(\pm 1)^n$ 通项不趋于 0. 发散.

$\therefore \frac{e^{-x}}{1-x} = 1+\sum_{n=2}^{\infty}\left(\sum_{k=2}^{n}\frac{(-1)^k}{k!}\right)x^n$. 收敛域 $(-1,1)$.

7. 利用幂级数展开, 计算下列积分, 要求精确到 0.001:

(1) $\int_0^1 \frac{\sin x}{x}\,dx$. (利用 6(9) 的结论)

解: $\int_0^1 \frac{\sin x}{x}\,dx = \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2\cdot(2n)!} = 1-\frac{1}{9\times 2!}+\frac{1}{25\times 4!}-\frac{1}{49\times 6!}+\cdots$

$\doteq 1-\frac{1}{18}+\frac{1}{600}-\frac{1}{35280}+\cdots = 1-0.05556+0.00169$

≈ 0.946

(2) $\int_0^1 \cos x^2 dx$. $\left(\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}\right)$

解: $\int_0^1 \cos x^2 dx = \int_0^1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{4n} dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \int_0^1 x^{4n} dx$

$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \cdot \frac{1}{4n+1} = 1 - \frac{1}{5\times 2!} + \frac{1}{9\times 4!} - \frac{1}{13\times 6!} + \cdots$

$= 1 - \frac{1}{10} + \frac{1}{216} - \frac{1}{9360} + \cdots = 1 - 0.1 + 0.00463 - \cdots$

≈ 0.905.

(3) $\int_0^{\frac{1}{2}} \frac{\arctan x}{x} dx$. $\left(\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}\right)$

解: $\int_0^{\frac{1}{2}} \frac{\arctan x}{x} dx = \int_0^{\frac{1}{2}} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n} dx = \sum_{n=0}^{\infty} \int_0^{\frac{1}{2}} \frac{(-1)^n}{2n+1} x^{2n} dx$

$= \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \cdot \frac{1}{2n+1} x^{2n+1} \Big|_0^{\frac{1}{2}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2} \cdot \left(\frac{1}{2}\right)^{2n+1}$

$= \frac{1}{1\times 2} - \frac{1}{9\times 2^3} + \frac{1}{25\times 2^5} - \frac{1}{49\times 2^7} + \frac{1}{81\times 2^9} - \cdots$

$= \frac{1}{2} - \frac{1}{72} + \frac{1}{800} - \frac{1}{6272} + \frac{1}{41472} - \cdots \approx 0.487$.

8. 将函数 $\ln(x+\sqrt{1+x^2})$ 展开成 Maclaurin 级数.

解: 设 $f(x) = \ln(x+\sqrt{1+x^2})$. $f'(x) = \frac{1}{x+\sqrt{1+x^2}} \cdot \left(1 + \frac{x}{\sqrt{1+x^2}}\right) = \frac{1}{\sqrt{1+x^2}} = (1+x^2)^{-\frac{1}{2}}$.

$f'(x) = \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} x^{2n}$.

$f(x) = \int_0^x \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} t^{2n} dt = \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} \int_0^x t^{2n} dt = \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} \frac{x^{2n+1}}{2n+1}$.

$\therefore \ln\{x+\sqrt{1+x^2}\} = \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} \frac{x^{2n+1}}{2n+1}$, $x \in (-1,1)$.

9. 求幂级数 $\sum_{n=1}^{\infty} n 2^{\frac{n}{2}} x^{3n-1}$ 的和函数.

解: $\sum_{n=1}^{\infty} n \cdot 2^{\frac{n}{2}} x^{3n-1} = \sqrt{2} x^2 \sum_{n=1}^{\infty} n \left(\sqrt{2} x^3\right)^{n-1} = \sqrt{2} x^2 \cdot \frac{1}{(1-\sqrt{2}x^3)^2}$ $\left(\sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2}\right)$

$= \frac{\sqrt{2} x^2}{(1-\sqrt{2}x^3)^2}$. 当 $|\sqrt{2}x^3| < 1$ 即: $x \in \left(-\frac{1}{\sqrt[3]{2}}, \frac{1}{\sqrt[3]{2}}\right)$ 时成立.

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10. 利用 $\frac{e^x-1}{x}$ 的幂级数展开，证明：$\sum_{n=1}^{\infty}\frac{n}{(n+1)!}=1$.

证明：设 $f(x)=\frac{e^x-1}{x}=\frac{1}{x}\left(\sum_{n=0}^{\infty}\frac{x^n}{n!}-1\right)=\frac{1}{x}\left(\sum_{n=1}^{\infty}\frac{x^n}{n!}\right)=\sum_{n=1}^{\infty}\frac{x^{n-1}}{n!}$

$=\sum_{n=0}^{\infty}\frac{x^n}{(n+1)!}$. $f'(x)=\sum_{n=1}^{\infty}\frac{nx^{n-1}}{(n+1)!}\Rightarrow f'(1)=\sum_{n=1}^{\infty}\frac{n}{(n+1)!}$

$\because f(x)=\frac{e^x-1}{x}$. $f'(x)=\frac{xe^x-(e^x-1)}{x^2}=\frac{(x-1)e^x+1}{x^2}\Rightarrow f'(1)=1$.

$\therefore \sum_{n=1}^{\infty}\frac{n}{(n+1)!}=1$.

11. 设 $I_n=\int_0^{\frac{\pi}{4}}\sin^n x\cos x\,dx$, $n=0,1,2,\cdots$, 求 $\sum_{n=1}^{\infty}I_n$.

解：$I_n=\int_0^{\frac{\pi}{4}}\sin^n x\cos x\,dx=\int_0^{\frac{\pi}{4}}\sin^n x\,d\sin x=\frac{1}{n+1}\sin^{(n+1)}x\Big|_0^{\frac{\pi}{4}}=\frac{1}{n+1}\cdot\left(\frac{1}{\sqrt{2}}\right)^{n+1}$.

$\sum_{n=1}^{\infty}I_n=\sum_{n=1}^{\infty}\frac{1}{n+1}\cdot\left(\frac{1}{\sqrt{2}}\right)^{n+1}$. 而 $\sum_{n=1}^{\infty}\frac{1}{n+1}x^{n+1}=\int_0^x\sum_{n=1}^{\infty}t^n\,dt$

$=\int_0^x\left(\frac{1}{1-t}-1\right)dt=\ln\frac{1}{1-x}-x$.

$\therefore \sum_{n=1}^{\infty}I_n=\sum_{n=1}^{\infty}\frac{1}{n+1}\left(\frac{1}{\sqrt{2}}\right)^{n+1}=\ln\frac{1}{1-\frac{1}{\sqrt{2}}}-\frac{1}{\sqrt{2}}=\ln(2+\sqrt{2})-\frac{1}{\sqrt{2}}\approx 0.5208$.

§3. Fourier 级数

习题 P.216

1. 设 f 是以 2π 为周期的函数，在 $[-\pi,\pi)$ 上的表达式如下，将 f 展开成 Fourier 级数.

(1) $f(x)=\begin{cases}x, & x\in[-\pi,0)\\ 0, & x\in[0,\pi)\end{cases}$;

解：$a_0=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\,dx=\frac{1}{\pi}\int_{-\pi}^{0}x\,dx=\frac{1}{\pi}\cdot\frac{1}{2}x^2\Big|_{-\pi}^{0}=-\frac{\pi}{2}$.

$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\cos nx\,dx=\frac{1}{\pi}\int_{-\pi}^{0}x\cos nx\,dx=\frac{1}{n\pi}\int_{-\pi}^{0}x\,d\sin nx$

$=\frac{1}{n\pi}x\sin nx\Big|_{-\pi}^{0}-\frac{1}{n\pi}\int_{-\pi}^{0}\sin nx\,dx=\frac{1}{n^2\pi}\cos nx\Big|_{-\pi}^{0}=\frac{1-(-1)^n}{n^2\pi}$.

$b_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\sin nx\,dx=\frac{1}{\pi}\int_{-\pi}^{0}x\sin nx\,dx=-\frac{1}{n\pi}\int_{-\pi}^{0}x\,d\cos nx$

$$= -\frac{1}{n\pi} x\cos nx \Big|_{-\pi}^{0} + \frac{1}{n\pi}\int_{-\pi}^{0} \cos nx\,dx = \frac{(-1)^n}{n\pi}$$

$$\therefore f(x) \sim -\frac{\pi}{4} + \sum_{n=1}^{\infty}\left[\frac{1-(-1)^n}{n^2\pi}\cos nx + \frac{(-1)^n}{n\pi}\sin nx\right] \quad x\in[-\pi,\pi).$$

(2) $f(x) = |\cos x|$.

解: $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi}|\cos x|\,dx = \frac{1}{\pi}\int_{-\pi}^{-\frac{\pi}{2}}(-\cos x)\,dx + \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos x\,dx + \frac{1}{\pi}\int_{\frac{\pi}{2}}^{\pi}(-\cos x)\,dx$

$$= \frac{1}{\pi}(-\sin x)\Big|_{-\pi}^{-\frac{\pi}{2}} + \frac{1}{\pi}\sin x\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \frac{1}{\pi}(-\sin x)\Big|_{\frac{\pi}{2}}^{\pi} = \frac{1}{\pi}+\frac{2}{\pi}+\frac{1}{\pi} = \frac{4}{\pi}.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\left(\int_{-\pi}^{-\frac{\pi}{2}} + \int_{\frac{\pi}{2}}^{\pi}\right)(-\cos x\cos nx)\,dx + \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos x\cos nx\,dx$$

$$= -\frac{1}{2\pi}\left(\int_{-\pi}^{-\frac{\pi}{2}} + \int_{\frac{\pi}{2}}^{\pi}\right)\left[\cos(n+1)x + \cos(n-1)x\right]dx + \frac{1}{2\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[\cos(n+1)x + \cos(n-1)x\right]dx$$

$$= -\frac{1}{2\pi}\left[\frac{\sin(n+1)x}{n+1}\Big|_{-\pi}^{-\frac{\pi}{2}} + \frac{\sin(n+1)x}{n+1}\Big|_{\frac{\pi}{2}}^{\pi} + \frac{\sin(n-1)x}{n-1}\Big|_{-\pi}^{-\frac{\pi}{2}} + \frac{\sin(n-1)x}{n-1}\Big|_{\frac{\pi}{2}}^{\pi}\right]$$

$$+\frac{1}{2\pi}\left[\frac{\sin(n+1)x}{n+1}\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \frac{\sin(n-1)x}{n-1}\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\right]$$

$$= \frac{1}{\pi}\left[\frac{\sin\frac{n+1}{2}\pi}{n+1} + \frac{\sin\frac{n-1}{2}\pi}{n-1}\right] + \frac{1}{\pi}\left[-\frac{\sin\frac{n+1}{2}\pi}{n+1} + \frac{\sin\frac{n-1}{2}\pi}{n-1}\right]$$

$$= \frac{2}{\pi}\cdot\frac{1}{n^2-1}\left[(n-1)\sin\left(\frac{n\pi}{2}+\frac{\pi}{2}\right) + (n+1)\sin\left(\frac{n\pi}{2}-\frac{\pi}{2}\right)\right] = -\frac{4}{\pi(n^2-1)}\cos\frac{n\pi}{2}.$$

$\because f(-x) = f(x)$. 为偶函数 $\therefore b_n = 0$.

$$\therefore f(x) \sim \frac{2}{\pi} - \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\cos\frac{n\pi}{2}}{n^2-1}\cos nx. \qquad x\in[-\pi,\pi).$$

(3) $f(x) = \begin{cases} ax & x\in[-\pi,0) \\ bx & x\in[0,\pi) \end{cases}$

解: $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\int_{-\pi}^{0} ax\,dx + \frac{1}{\pi}\int_{0}^{\pi} bx\,dx = \frac{a}{2\pi}x^2\Big|_{-\pi}^{0} + \frac{b}{2\pi}x^2\Big|_{0}^{\pi}$

$$= -\frac{\pi a}{2} + \frac{\pi b}{2} = \frac{b-a}{2}\pi.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_{-\pi}^{0} ax\cos nx\,dx + \frac{1}{\pi}\int_{0}^{\pi} bx\cos nx\,dx$$

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$$= \frac{a}{n\pi}\int_{-\pi}^{0} x\,d\sin nx + \frac{b}{n\pi}\int_{0}^{\pi} x\,d\sin nx$$

$$= -\frac{a}{n\pi}\int_{-\pi}^{0} \sin nx\,dx - \frac{b}{n\pi}\int_{0}^{\pi} \sin nx\,dx$$

$$= -\frac{a}{n^2\pi}\cos nx\Big|_{-\pi}^{0} + \frac{b}{n^2\pi}\cos nx\Big|_{0}^{\pi} = -\frac{a}{n^2\pi}\left(1-(-1)^n\right) + \frac{b}{n^2\pi}\left((-1)^n-1\right)$$

$$= \frac{2(a-b)}{\pi}\cdot\frac{1}{(2k+1)^2} \qquad n=2k+1.$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\int_{-\pi}^{0} ax\sin nx\,dx + \frac{1}{\pi}\int_{0}^{\pi} bx\sin nx\,dx$$

$$= -\frac{a}{n\pi}\int_{-\pi}^{0} x\,d\cos nx - \frac{b}{n\pi}\int_{0}^{\pi} x\,d\cos nx$$

$$= -\frac{a}{n\pi}x\cos nx\Big|_{-\pi}^{0} - \frac{b}{n\pi}x\cos nx\Big|_{0}^{\pi} = -\frac{a}{n}(-1)^n - \frac{b}{n}(-1)^n$$

$$= \frac{a+b}{n}(-1)^{n+1}$$

$$\therefore f(x) \sim \frac{b-a}{4}\pi + \frac{2(a-b)}{\pi}\sum_{n=0}^{\infty}\frac{\cos(2n+1)x}{(2n+1)^2} + (a+b)\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin nx.$$

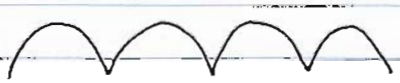
$x \in (-\pi, \pi)$.

2. 设交流电的变化规律为 $E(t) = A\sin\omega t$，它转变为直流电的整流过程有两种类型：

a) 半波整流 $f_1(t) = \frac{A}{2}(\sin\omega t + |\sin\omega t|)$



b) 全波整流 $f_2(t) = A|\sin\omega t|$



现取 $\omega = 1$，试将 $f_1(t)$ 和 $f_2(t)$ 在 $[-\pi, \pi]$ 展开 Fourier 级数.

解：a) $f_1(t) = \frac{A}{2}(\sin t + |\sin t|)$ $t \in [-\pi, \pi]$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi}\frac{A}{2}(\sin t + |\sin t|)\,dt = \frac{1}{\pi}\int_{0}^{\pi}\frac{A}{2}\cdot 2\sin t\,dt = -\frac{A}{\pi}\cos t\Big|_{0}^{\pi} = \frac{2A}{\pi}.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_{0}^{\pi} A\sin t\cos nt\,dt = \frac{A}{2\pi}\int_{0}^{\pi}\left[\sin(1+n)t + \sin(1-n)t\right]dt$$

$$= \frac{A}{2\pi}\left[-\frac{\cos(n+1)t}{n+1}\Big|_{0}^{\pi} + \frac{\cos(n-1)t}{n-1}\Big|_{0}^{\pi}\right] = \frac{A}{2\pi}\left[\frac{1-(-1)^{n+1}}{n+1} + \frac{(-1)^{n-1}-1}{n-1}\right]$$

$$= \frac{A}{2\pi}\left[\frac{1-(-1)^{n+1}}{n+1}+\frac{1-(-1)^{n+1}}{1-n}\right] = \frac{A}{\pi}\cdot\frac{1-(-1)^{n+1}}{1-n^2} = \frac{A}{\pi}\cdot\frac{2}{1-4k^2}. \quad (n=2k).$$

$$b_1 = \frac{1}{\pi}\int_{-\pi}^{\pi} f_1(t)\cdot\sin t\,dt = \frac{1}{\pi}\int_0^{\pi} A\sin^2 t\,dt = \frac{A}{2\pi}\int_0^{\pi}(1-\cos 2t)\,dt = \frac{A}{2}.$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f_1(t)\cdot\sin nt\,dt = \frac{1}{\pi}\int_0^{\pi} A\sin t\sin nt\,dt = \frac{A}{2\pi}\int_0^{\pi}\left[\cos(n+1)t-\cos(n+1)t\right]dt = 0.$$

$(n=2,3,\cdots)$.

$$\therefore f_1(t) \sim \frac{A}{\pi} + \frac{A}{\pi}\sum_{k=1}^{\infty}\frac{2}{1-4k^2}\cos 2kt + \frac{A}{2}\sin t.$$

(2) $f_2(t) = A\left|\sin\omega t\right|$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f_2(t)\,dt = \frac{2}{\pi}\int_0^{\pi} A\sin t\,dt = \frac{2A}{\pi}(-\cos t)\Big|_0^{\pi} = \frac{4A}{\pi}.$$

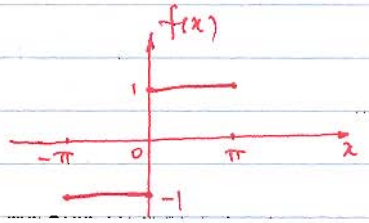
$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f_2(t)\cos nt\,dt = \frac{1}{\pi}\int_0^{\pi} A\sin t\cos nt\,dt = \frac{2A}{\pi}\int_0^{\pi}\sin t\cos nt\,dt$$

$$= \frac{A}{\pi}\cdot\frac{4}{1-4k^2}. \quad (n=2k). \qquad \because f_2(-t)=f_2(t). \text{ 为偶函数}. \therefore b_n = 0.$$

$$\therefore f_2(t) \sim \frac{2A}{\pi} + \frac{A}{\pi}\sum_{k=1}^{\infty}\frac{4}{1-4k^2}\cos 2kt.$$

3. 将下列函数展开成正弦级数:

(1) $f(x)=1. \qquad x\in[0,\pi]$



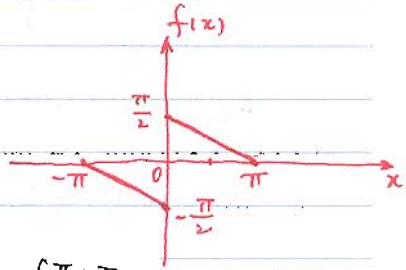
解: 将 $f(x)$ 延拓成奇函数: $f(x)=\begin{cases}1 & x\in[0,\pi]\\ -1 & x\in[-\pi,0).\end{cases}$

$$a_n = 0. \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\int_{-\pi}^{0}(-\sin nx)\,dx + \frac{1}{\pi}\int_0^{\pi}\sin nx\,dx$$

$$\left(b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx\right) = \frac{1}{n\pi}\cos nx\Big|_{-\pi}^{0} - \frac{1}{n\pi}\cos nx\Big|_0^{\pi} = \frac{1}{n\pi}\left[1-(-1)^n-(-1)^n+1\right] = \frac{2}{n\pi}\left[1-(-1)^n\right]$$

$$\therefore f(x) \sim \sum_{n=1}^{\infty}\frac{2}{n\pi}\left[1-(-1)^n\right]\sin nx = \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{1}{2n+1}\sin(2n+1)\pi.$$

(2) $f(x)=\frac{\pi-x}{2}. \qquad x\in[0,\pi)$



解: 将 $f(x)$ 延拓成奇函数: $f(x)=\begin{cases}\frac{\pi-x}{2} & x\in[0,\pi]\\ -\frac{x+\pi}{2} & x\in[-\pi,0).\end{cases}$

$$a_n = 0. \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\int_{-\pi}^{0}\left(-\frac{x+\pi}{2}\right)\sin nx\,dx + \frac{1}{\pi}\int_0^{\pi}\left(\frac{\pi-x}{2}\right)\sin nx\,dx$$

$$= -\frac{1}{n\pi}\int_{-\pi}^{0}\left(-\frac{x+\pi}{2}\right)d\cos nx - \frac{1}{n\pi}\int_{0}^{\pi}\frac{\pi-x}{2}\,d\cos nx$$

$$= \frac{x+\pi}{2n\pi}\cos nx\Big|_{-\pi}^{0} - \frac{\pi-x}{2n\pi}\cos nx\Big|_{0}^{\pi} = \frac{1}{2n} + \frac{1}{2n} = \frac{1}{n}. \quad \left(= \frac{2}{\pi}\int_0^{\pi}\frac{\pi-x}{2}\sin nx\,dx\right)$$

$$\therefore f(x) \sim \sum_{n=1}^{\infty}\frac{1}{n}\sin nx.$$

(3) $f(x) = \begin{cases} 2x, & x\in[0,\frac{\pi}{2}) \\ \pi, & x\in(\frac{\pi}{2},\pi). \end{cases}$

解: $a_n = 0$. $b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx = \frac{2}{\pi}\int_0^{\frac{\pi}{2}} 2x\sin nx\,dx + \frac{2}{\pi}\int_{\frac{\pi}{2}}^{\pi}\pi\sin nx\,dx$

$$= -\frac{4}{n\pi}\int_0^{\frac{\pi}{2}} x\,d\cos nx - \frac{2}{n}\int_{\frac{\pi}{2}}^{\pi} d\cos nx$$

$$= -\frac{4}{n\pi}x\cos nx\Big|_0^{\frac{\pi}{2}} + \frac{4}{n\pi}\int_0^{\frac{\pi}{2}}\cos nx\,dx - \frac{2}{n}\cos nx\Big|_{\frac{\pi}{2}}^{\pi}$$

$$= -\frac{4}{n\pi}\cdot\frac{\pi}{2}\cos\frac{n\pi}{2} + \frac{4}{n^2\pi}\sin nx\Big|_0^{\frac{\pi}{2}} - \frac{2}{n}\cos nx\Big|_{\frac{\pi}{2}}^{\pi}$$

$$= -\frac{2}{n}\cos\frac{n\pi}{2} + \frac{4}{n^2\pi}\sin\frac{n\pi}{2} - \frac{2}{n}(-1)^n + \frac{2}{n}\cos\frac{n\pi}{2}$$

$$= \frac{2}{n}(-1)^{n+1} + \frac{4}{n^2\pi}\sin\frac{n\pi}{2}.$$

$$\therefore f(x) \sim \sum_{n=1}^{\infty}\left[\frac{2}{n}(-1)^{n+1} + \frac{4}{n^2\pi}\sin\frac{n\pi}{2}\right]\sin nx.$$

(4) $f(x) = \begin{cases} \cos\frac{\pi x}{2}, & x\in[0,1) \\ 0, & x\in[1,2) \end{cases}$ (周期 $2T=4 \Rightarrow T=2$ (半周期))

解: $a_n = 0$. $b_n = \frac{2}{T}\int_0^{T} f(x)\sin\frac{n\pi}{T}x\,dx = \int_0^{2} f(x)\sin\frac{n\pi}{2}x\,dx$

$$= \int_0^1 \cos\frac{\pi x}{2}\sin\frac{n\pi x}{2}\,dx = \frac{1}{2}\int_0^1\left(\sin\frac{n+1}{2}\pi x + \sin\frac{n-1}{2}\pi x\right)dx$$

$$b_1 = \frac{1}{2}\int_0^1 \sin\pi x\,dx = -\frac{1}{2\pi}\cos\pi x\Big|_0^1 = \frac{1}{\pi}.$$

$$b_n = \frac{1}{2}\int_0^1 \sin\frac{n+1}{2}\pi x\,dx + \frac{1}{2}\int_0^1 \sin\frac{n-1}{2}\pi x\,dx$$

$$= -\frac{1}{(n+1)\pi}\cos\frac{n+1}{2}\pi x\Big|_0^1 - \frac{1}{(n-1)\pi}\cos\frac{n-1}{2}\pi x\Big|_0^1$$

$$= -\frac{1}{(n+1)\pi}\cos\left(\frac{n\pi}{2}+\frac{\pi}{2}\right) - \frac{1}{(n-1)\pi}\cos\left(\frac{n\pi}{2}-\frac{\pi}{2}\right) + \frac{1}{(n+1)\pi} + \frac{1}{(n-1)\pi}$$

$$= \frac{2}{\pi}\left[\frac{n}{n^2-1} - \frac{1}{n^2-1}\sin\frac{n\pi}{2}\right].$$

$$\therefore f(x) \sim \frac{1}{\pi}\sin\frac{\pi x}{2} + \frac{2}{\pi}\sum_{n=2}^{\infty}\frac{n-\sin\frac{n\pi}{2}}{n^2-1}\sin\frac{n\pi}{2}x.$$

4. 将下列函数展开成余弦级数：

(1) $f(x) = x+1, \quad x \in [0, \pi)$.

解：$b_n = 0$. $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx = \frac{2}{\pi}\int_0^{\pi}(x+1)\,dx = \frac{2}{\pi}\left(\frac{1}{2}x^2+x\right)\Big|_0^{\pi} = \pi+2$.

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_0^{\pi}(x+1)\cos nx\,dx = \frac{2}{n\pi}\int_0^{\pi}(x+1)\,d\sin nx$$

$$= \frac{2}{n\pi}(x+1)\sin nx\Big|_0^{\pi} - \frac{2}{n\pi}\int_0^{\pi}\sin nx\,dx = \frac{2}{n^2\pi}\cos nx\Big|_0^{\pi} = \frac{2}{n^2\pi}\left[(-1)^n-1\right].$$

$$\therefore f(x) \sim \frac{\pi+2}{2} + \sum_{n=1}^{\infty}\frac{2}{n^2\pi}\left[(-1)^n-1\right]\cos nx$$

$$= \frac{\pi+2}{2} - \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\cos(2n-1)x}{(2n-1)^2}$$

(2) $f(x) = x(\pi-x), \quad x \in [0, \pi]$

解：$b_n = 0$. $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx = \frac{2}{\pi}\int_0^{\pi} x(\pi-x)\,dx = \frac{2}{\pi}\left(\frac{\pi}{2}x^2 - \frac{1}{3}x^3\right)\Big|_0^{\pi} = \frac{\pi^2}{3}$.

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_0^{\pi} x(\pi-x)\cos nx\,dx$$

$$= \frac{2}{n\pi}x(\pi-x)\sin nx\Big|_0^{\pi} - \frac{2}{n\pi}\int_0^{\pi}(\pi-2x)\sin nx\,dx$$

$$= \frac{2}{n^2\pi}\int_0^{\pi}(\pi-2x)\,d\cos nx = \frac{2}{n^2\pi}(\pi-2x)\cos nx\Big|_0^{\pi} + \frac{4}{n^2\pi}\int_0^{\pi}\cos nx\,dx$$

$$= -\frac{2}{n^2}\left[1+(-1)^n\right]$$

$$\therefore f(x) \sim \frac{\pi^2}{6} + \sum_{n=1}^{\infty}\left[-\frac{2}{n^2}\left[1+(-1)^n\right]\right]\cos nx = \frac{\pi^2}{6} - \sum_{n=1}^{\infty}\frac{\cos 2nx}{n^2}$$

(3) $f(x) = \begin{cases} \sin 2x & x \in [0, \frac{\pi}{4}) \\ 1 & x \in [\frac{\pi}{4}, \frac{\pi}{2}) \end{cases}$.

($2T = \pi \Rightarrow T = \frac{\pi}{2}$（半周期））

解: $b_n=0$. $a_0=\frac{2}{T}\int_0^T f(x)dx=\frac{4}{\pi}\int_0^{\frac{\pi}{2}} f(x)dx=\frac{4}{\pi}\int_0^{\frac{\pi}{4}}\sin 2x\,dx+\frac{4}{\pi}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}1\,dx$

$=\frac{4}{\pi}\left(-\frac{1}{2}\cos 2x\Big|_0^{\frac{\pi}{4}}+x\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}\right)=\frac{4}{\pi}\left(\frac{1}{2}+\frac{\pi}{4}\right)=\frac{2+\pi}{\pi}$.

$a_n=\frac{2}{T}\int_0^T f(x)\cos\frac{n\pi}{T}x\,dx=\frac{4}{\pi}\int_0^{\frac{\pi}{2}} f(x)\cos 2nx\,dx$

$=\frac{4}{\pi}\int_0^{\frac{\pi}{4}}\sin 2x\cos 2nx\,dx+\frac{4}{\pi}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\cos 2nx\,dx$

$=\frac{2}{\pi}\int_0^{\frac{\pi}{4}}\left[\sin 2(n+1)x-\sin 2(n-1)x\right]dx+\frac{4}{\pi}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\cos 2nx\,dx$

$=\frac{2}{\pi}\left[-\frac{\cos 2(n+1)x}{2(n+1)}\Big|_0^{\frac{\pi}{4}}+\frac{\cos 2(n-1)x}{2(n-1)}\Big|_0^{\frac{\pi}{4}}\right]+\frac{4}{\pi}\cdot\frac{\sin 2nx}{2n}\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}$

$=\frac{2}{\pi n(n^2-1)}\left(\sin\frac{n\pi}{2}-n\right)$ ——$n=2,3,4,\cdots$

$a_1=\frac{2}{T}\int_0^T f(x)\cos 2x\,dx=\frac{4}{\pi}\int_0^{\frac{\pi}{4}}\sin 2x\cos 2x\,dx+\frac{4}{\pi}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\cos 2x\,dx=-\frac{2}{\pi}$

$\therefore f(x)\sim\frac{2+\pi}{2\pi}-\frac{2}{\pi}\cos 2x+\frac{2}{\pi}\sum_{n=2}^{\infty}\frac{1}{n(n^2-1)}\left(\sin\frac{n\pi}{2}-n\right)\cos 2nx$.

(4) $f(x)=e^x$, $x\in[0,\pi)$.

解: $b_n=0$, $a_0=\frac{2}{\pi}\int_0^{\pi}f(x)dx=\frac{2}{\pi}\int_0^{\pi}e^x dx=\frac{2}{\pi}e^x\Big|_0^{\pi}=\frac{2}{\pi}(e^{\pi}-1)$.

$a_n=\frac{2}{\pi}\int_0^{\pi}f(x)\cos nx\,dx=\frac{2}{\pi}\int_0^{\pi}e^x\cos nx\,dx=\frac{2}{n\pi}\int_0^{\pi}e^x\,d\sin nx$

$=\frac{2}{n\pi}e^x\sin nx\Big|_0^{\pi}-\frac{2}{n\pi}\int_0^{\pi}e^x\sin nx\,dx=\frac{2}{n^2\pi}\int_0^{\pi}e^x\,d\cos nx$

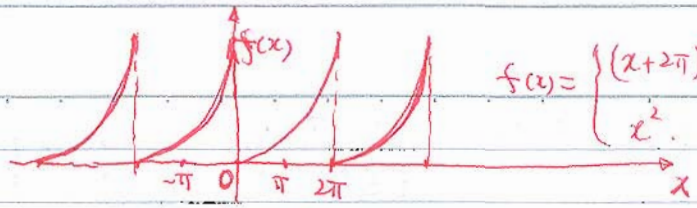
$=\frac{2}{n^2\pi}e^x\cos nx\Big|_0^{\pi}-\frac{2}{n^2\pi}\int_0^{\pi}e^x\cos nx\,dx=\frac{2}{n^2\pi}\left[(-1)^n e^{\pi}-1\right]-\frac{1}{n^2}a_n$.

$a_n=\frac{1}{1+\frac{1}{n^2}}\cdot\frac{2}{n^2\pi}\left[(-1)^n e^{\pi}-1\right]=\frac{2}{\pi(1+n^2)}\left[(-1)^n e^{\pi}-1\right]$.

$\therefore f(x)\sim\frac{1}{\pi}(e^{\pi}-1)+\sum_{n=1}^{\infty}\frac{2}{\pi(1+n^2)}\left[(-1)^n e^{\pi}-1\right]\cos nx$.

5、将下列函数在指定区间展开成 Fourier 级数：

11) $f(x) = x^2, \quad x \in [0, 2\pi]$.



$f(x) = \begin{cases} (x+2\pi)^2, & x \in [-\pi, 0) \\ x^2, & x \in [0, \pi) \end{cases}$

解:

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx = \frac{1}{\pi}\int_0^{2\pi} x^2\,dx = \frac{1}{\pi}\cdot\frac{1}{3}x^3\Big|_0^{2\pi} = \frac{8}{3}\pi^2.$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_0^{2\pi} x^2\cos nx\,dx = \frac{1}{n\pi}\int_0^{2\pi} x^2\,d\sin nx$$

$$= -\frac{2}{n\pi}\int_0^{2\pi} x\sin nx\,dx = \frac{2}{n^2\pi}x\cos nx\Big|_0^{2\pi} - \frac{2}{n^2\pi}\int_0^{2\pi}\cos nx\,dx = \frac{4}{n^2}.$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\int_0^{2\pi} x^2\sin nx\,dx = -\frac{1}{n\pi}\int_0^{2\pi} x^2\,d\cos nx$$

$$= -\frac{1}{n\pi}x^2\cos nx\Big|_0^{2\pi} + \frac{2}{n\pi}\int_0^{2\pi} x\cos nx\,dx = -\frac{4\pi}{n} + \frac{2}{n^2\pi}\int_0^{2\pi} x\,d\sin nx$$

$$= -\frac{4\pi}{n} - \frac{2}{n^2\pi}\int_0^{2\pi}\sin nx\,dx = -\frac{4\pi}{n}.$$

$$\therefore f(x) \sim \frac{4}{3}\pi^2 + \sum_{n=1}^{\infty}\left(\frac{4}{n^2}\cos nx - \frac{4\pi}{n}\sin nx\right).$$

12) $f(x) = \begin{cases} 2x+1, & x \in [-3, 0) \\ 1, & x \in [0, 3) \end{cases}$ $\quad (2T = 6 \Rightarrow T = 3)$

解:

$$a_0 = \frac{1}{T}\int_{-3}^{3} f(x)\,dx = \frac{1}{3}\int_{-3}^{0}(2x+1)\,dx + \frac{1}{3}\int_0^3 1\,dx = \frac{1}{3}(x^2+x)\Big|_{-3}^{0} + \frac{1}{3}x\Big|_0^3 = -1.$$

$$a_n = \frac{1}{T}\int_{-3}^{3} f(x)\cos\frac{n\pi}{T}x\,dx = \frac{1}{3}\int_{-3}^{0}(2x+1)\cos\frac{n\pi}{3}x\,dx + \frac{1}{3}\int_0^3 1\cdot\cos\frac{n\pi}{3}x\,dx$$

$$= \frac{1}{n\pi}\int_{-3}^{0}(2x+1)\,d\sin\frac{n\pi}{3}x + \frac{1}{n\pi}\int_0^3 d\sin\frac{n\pi}{3}x$$

$$= -\frac{2}{n\pi}\int_{-3}^{0}\sin\frac{n\pi}{3}x\,dx = \frac{6}{n^2\pi^2}\left[1-(-1)^n\right].$$

$$b_n = \frac{1}{T}\int_{-T}^{T} f(x)\sin\frac{n\pi}{T}x\,dx = \frac{1}{3}\int_{-3}^{0}(2x+1)\sin\frac{n\pi}{3}x\,dx + \frac{1}{3}\int_0^3\sin\frac{n\pi}{3}x\,dx$$

$$= -\frac{1}{n\pi}(2x+1)\cos\frac{n\pi}{3}x\Big|_{-3}^{0} + \frac{2}{n\pi}\int_{-3}^{0}\cos\frac{n\pi}{3}x\,dx - \frac{1}{n\pi}\cos\frac{n\pi}{3}x\Big|_0^3$$

$$= -\frac{1}{n\pi} - \frac{5}{n\pi}(-1)^n - \frac{1}{n\pi}(-1)^n + \frac{1}{n\pi} = -\frac{6}{n\pi}(-1)^n = \frac{6}{n\pi}(-1)^{n+1}$$

$$\therefore f(x) \sim -\frac{1}{2} + \sum_{n=1}^{\infty}\left\{\frac{6}{n^2\pi^2}\left[1-(-1)^n\right]\cos\frac{n\pi}{3}x + \frac{6}{n\pi}(-1)^{n+1}\sin\frac{n\pi}{3}x\right\}.$$

(3) $f(x)=\begin{cases} C & x\in[-T,0) \\ 0 & x\in[0,T) \end{cases}$, ($C$ 为常数).

解: $a_0=\frac{1}{T}\int_{-T}^{T} f(x)\,dx=\frac{1}{T}\int_{-T}^{0} C\,dx=\frac{1}{T}\cdot Cx\Big|_{-T}^{0}=C$.

$a_n=\frac{1}{T}\int_{-T}^{T} f(x)\cos\frac{n\pi}{T}x\,dx=\frac{1}{T}\int_{-T}^{0} C\cdot\cos\frac{n\pi}{T}x\,dx=\frac{C}{n\pi}\sin\frac{n\pi}{T}x\Big|_{-T}^{0}=0$.

$b_n=\frac{1}{T}\int_{-T}^{T} f(x)\sin\frac{n\pi}{T}x\,dx=\frac{1}{T}\int_{-T}^{0} C\cdot\sin\frac{n\pi}{T}x\,dx=-\frac{C}{n\pi}\cos\frac{n\pi}{T}x\Big|_{-T}^{0}$

$=\frac{C}{n\pi}\left[(-1)^n-1\right]$.

$\therefore f(x)\sim\frac{C}{2}+\sum_{n=1}^{\infty}\frac{C}{n\pi}\left[(-1)^n-1\right]\sin\frac{n\pi}{T}x=\frac{C}{2}-\frac{2C}{\pi}\sum_{n=1}^{\infty}\frac{1}{2n-1}\sin\frac{(2n-1)\pi}{T}x$.

(4) $f(x)=\begin{cases} e^{3x} & x\in[-1,0) \\ 0 & x\in[0,1) \end{cases}$ $(2T=2\Rightarrow T=1)$

解: $a_0=\frac{1}{T}\int_{-T}^{T} f(x)\,dx=\int_{-1}^{0} e^{3x}\,dx=\frac{1}{3}e^{3x}\Big|_{-1}^{0}=\frac{1}{3}(1-e^{-3})$.

$a_n=\frac{1}{T}\int_{-T}^{T} f(x)\cos\frac{n\pi}{T}x\,dx=\int_{-1}^{0} e^{3x}\cos n\pi x\,dx=\frac{1}{3}\int_{-1}^{0}\cos n\pi x\,de^{3x}$

$=\frac{1}{3}e^{3x}\cos n\pi x\Big|_{-1}^{0}+\frac{n\pi}{3}\int_{-1}^{0} e^{3x}\sin n\pi x\,dx$

$=\frac{1}{3}-\frac{1}{3}e^{-3}(-1)^n+\frac{n\pi}{9}\int_{-1}^{0}\sin n\pi x\,de^{3x}$

$=\frac{1}{3}-\frac{(-1)^n}{3e^3}+\frac{n\pi}{9}e^{3x}\sin n\pi x\Big|_{-1}^{0}-\frac{n^2\pi^2}{9}\int_{-1}^{0} e^{3x}\cos n\pi x\,dx$

$=\frac{1}{3}-\frac{(-1)^n}{3e^3}-\frac{n^2\pi^2}{9}a_n$. $\therefore a_n=\frac{3}{9+n^2\pi^2}\left[1-(-1)^n e^{-3}\right]$.

$b_n=\frac{1}{T}\int_{-T}^{T} f(x)\sin\frac{n\pi}{T}x\,dx=\int_{-1}^{0} e^{3x}\sin n\pi x\,dx=\frac{1}{3}\int_{-1}^{0}\sin n\pi x\,de^{3x}$

$=-\frac{n\pi}{3}\int_{-1}^{0} e^{3x}\cos n\pi x\,dx=-\frac{n\pi}{9}\int_{-1}^{0}\cos n\pi x\,de^{3x}$

$=-\frac{n\pi}{9}e^{3x}\cos n\pi x\Big|_{-1}^{0}-\frac{n^2\pi^2}{9}\int_{-1}^{0} e^{3x}\sin n\pi x\,dx$

$=-\frac{n\pi}{9}\left[1-(-1)^n e^{-3}\right]-\frac{n^2\pi^2}{9}b_n$. $\therefore b_n=\frac{n\pi}{9+n^2\pi^2}\left[-1+(-1)^n e^{-3}\right]$.

$$\therefore f(x)\sim\frac{1}{6}(1-e^{-3})+\sum_{n=1}^{\infty}\left\{\frac{3[1-(-1)^n e^{-3}]}{9+n^2\pi^2}\cos n\pi x+\frac{n\pi[-1+(-1)^n e^{-3}]}{9+n^2\pi^2}\sin n\pi x\right\}.$$

(5) $f(x)=10-x,\ x\in(5,15)$. $(2T=15-5=10.\Rightarrow T=5.)$

解: $a_0=\frac{1}{T}\int_5^{15}f(x)dx=\frac{1}{5}\int_5^{15}(10-x)dx=\frac{1}{5}\left(10x-\frac{1}{2}x^2\right)\Big|_5^{15}=0.$

$$a_n=\frac{1}{T}\int_5^{15}f(x)\cos\frac{n\pi}{T}x\,dx=\frac{1}{5}\int_5^{15}(10-x)\cos\frac{n\pi}{5}x\,dx$$

令 $x=5(t+2)$. 则:

$$a_n=\int_{-1}^{1}(-t)\cos n\pi t\,dt=\int_{-1}^{0}(-t)\cos n\pi t\,dt+\int_0^1(-t)\cos n\pi t\,dt$$

$$=-\int_0^1(-t)\cos n\pi t\,dt+\int_0^1(-t)\cos n\pi t\,dt=0.\quad(f(t)=-t\text{ 为奇函数})$$

$$b_n=\frac{1}{T}\int_5^{15}f(x)\sin\frac{n\pi}{T}x\,dx=\frac{1}{5}\int_5^{15}(10-x)\sin\frac{n\pi}{5}x\,dx\quad(\text{令 }x=5(t+2))$$

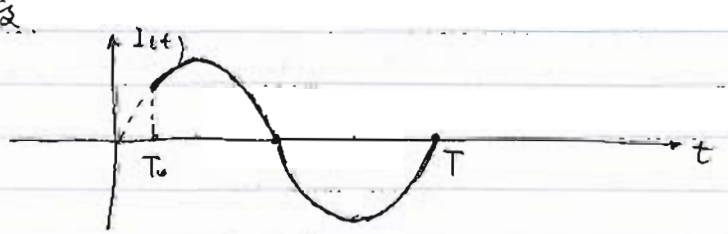
$$=\int_{-1}^{1}(-t)\sin n\pi t\,dt=2\int_0^1(-t)\sin n\pi t\,dt=\frac{2}{n\pi}\int_0^1 t\,d\cos n\pi t$$

$$=\frac{2}{n\pi}t\cos n\pi t\Big|_0^1-\frac{2}{n\pi}\int_0^1\cos n\pi t\,dt=\frac{2}{n\pi}(-1)^n.$$

$$\therefore f(x)\sim\sum_{n=1}^{\infty}\frac{2}{n\pi}(-1)^n\sin\frac{n\pi}{5}x.$$

b. 某可控硅控制电路中的负载电流为 $I(t)=\begin{cases}0 & 0\le t<T_0\\ 5\sin\omega t & T_0\le t<T\end{cases}$

其中 ω 为圆频率，周期 $T=\frac{2\pi}{\omega}$；现设初始导通时间 $T_0=\frac{T}{8}$，求 $I(t)$ 在 $[0,T]$ 上的 Fourier 级数.



解: $a_0=\frac{2}{T}\int_0^T I(t)dt=\frac{2}{T}\int_{T_0}^{T}5\sin\omega t\,dt=-\frac{10}{\omega T}\int_{T_0}^{T}d\cos\omega t$

$$=-\frac{5}{\pi}\cos\omega t\Big|_{\frac{T}{8}}^{T}=-\frac{5}{\pi}\left(1-\frac{\sqrt{2}}{2}\right)=-\frac{5(2-\sqrt{2})}{2\pi}=\frac{5(\sqrt{2}-2)}{2\pi}.$$

$a_n = \frac{2}{T}\int_0^T I(t)\cos\frac{2n\pi}{T}t\,dt = \frac{2}{T}\int_{T_0}^{T} 5\sin\omega t\cdot\cos\frac{2n\pi}{T}t\,dt = \frac{10}{T}\int_{T_0}^{T}\sin\omega t\cos n\omega t\,dt$

$= \frac{5}{\pi}\int_{\frac{T}{8}}^{T}\sin\omega t\cdot\cos n\omega t\,d\omega t = \frac{5}{\pi}\int_{\frac{\pi}{4}}^{2\pi}\sin x\cdot\cos nx\,dx. \quad (x=\omega t)$

$= \frac{5}{2\pi}\int_{\frac{\pi}{4}}^{2\pi}\left[\sin(n+1)x - \sin(n-1)x\right]dx = \frac{5}{2\pi}\left[\frac{1}{n+1}\cos\frac{n+1}{4}\pi - \frac{1}{n-1}\cos\frac{n-1}{4}\pi + \frac{2}{n^2-1}\right]$ $n=2,3,4,\cdots$

$a_1 = \frac{5}{\pi}\int_{\frac{\pi}{4}}^{2\pi}\sin x\cdot\cos x\,dx = \frac{5}{2\pi}\int_{\frac{\pi}{4}}^{2\pi}\sin 2x\,dx$

$= -\frac{5}{4\pi}\cos 2x\Big|_{\frac{\pi}{4}}^{2\pi} = -\frac{5}{4\pi}.$

$b_n = \frac{2}{T}\int_0^T I(t)\sin\frac{2n\pi}{T}t\,dt = \frac{2}{T}\int_{T_0}^{T} 5\sin\omega t\cdot\sin\frac{2n\pi}{T}t\,dt = \frac{10}{T}\int_{T_0}^{T}\sin\omega t\cdot\sin n\omega t\,dt$

$= \frac{5}{\pi}\int_{\frac{T}{8}}^{T}\sin\omega t\cdot\sin n\omega t\,d\omega t = \frac{5}{\pi}\int_{\frac{\pi}{4}}^{2\pi}\sin x\cdot\sin nx\,dx. \quad (x=\omega t)$

$= \frac{5}{2\pi}\int_{\frac{\pi}{4}}^{2\pi}\left[\cos(n+1)x - \cos(n-1)x\right]dx = \frac{5}{2\pi}\left[\frac{1}{n+1}\sin\frac{n+1}{4}\pi - \frac{1}{n-1}\sin\frac{n-1}{4}\pi\right]$ $n=2,3,4\cdots$

$b_1 = \frac{5}{\pi}\int_{\frac{\pi}{4}}^{2\pi}\sin x\cdot\sin x\,dx = \frac{5}{\pi}\int_{\frac{\pi}{4}}^{2\pi}\sin^2 x\,dx = \frac{5}{2\pi}\int_{\frac{\pi}{4}}^{2\pi}(1-\cos 2x)\,dx$

$= \frac{5}{2\pi}\left(x - \frac{1}{2}\sin 2x\right)\Big|_{\frac{\pi}{4}}^{2\pi} = \frac{5}{2\pi}\left(2\pi - \frac{\pi}{4} + \frac{1}{2}\right) = \frac{5(7\pi+2)}{8\pi}$

$\therefore I(t) \sim -\frac{5}{4\pi}(2-\sqrt{2}) - \frac{5}{4\pi}\cos\omega t + \left(\frac{5}{4\pi}+\frac{35}{8}\right)\sin\omega t$

$+\frac{5}{2\pi}\sum_{n=2}^{\infty}\left\{\left[\frac{1}{n+1}\cos\frac{n+1}{4}\pi - \frac{1}{n-1}\cos\frac{n-1}{4}\pi + \frac{2}{n^2-1}\right]\cos n\omega t\right.$

$\left.+\left[\frac{1}{n+1}\sin\frac{n+1}{4}\pi - \frac{1}{n-1}\sin\frac{n-1}{4}\pi\right]\sin n\omega t\right\}.$

7. 将 $f(x)=\begin{cases}0, & -\pi\le x\le 0\\ \sin x, & 0\le x\le\pi\end{cases}$ 展开为 Fourier 级数，并求级数 $\sum_{n=1}^{\infty}\frac{1}{1-4n^2}$ 的和.

解: $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\int_0^{\pi}\sin x\,dx = \frac{1}{\pi}(-\cos x)\Big|_0^{\pi} = \frac{2}{\pi}.$

$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_0^{\pi}\sin x\cos nx\,dx$

$= \frac{1}{2\pi}\int_0^{\pi}\left[\sin(n+1)x - \sin(n-1)x\right]dx = \frac{1}{2\pi}\left[\frac{1}{n-1}\cos(n-1)x - \frac{1}{n+1}\cos(n+1)x\right]\Big|_0^{\pi}$

$$= \frac{1}{2\pi}\left[\frac{1}{n+1}\cdot(-1)^{n+1} - \frac{1}{n+1}\cdot(-1)^{n+1} - \frac{1}{n-1} + \frac{1}{n+1}\right] = \frac{(-1)^{n+1}-1}{(n^2-1)\pi} \quad n=2,3,4,\cdots$$

$$a_1 = \frac{1}{\pi}\int_0^{\pi} \sin x\cos x\,dx = \frac{1}{2\pi}\int_0^{\pi}\sin 2x\,dx = -\frac{1}{4\pi}\cos 2x\Big|_0^{\pi} = 0.$$

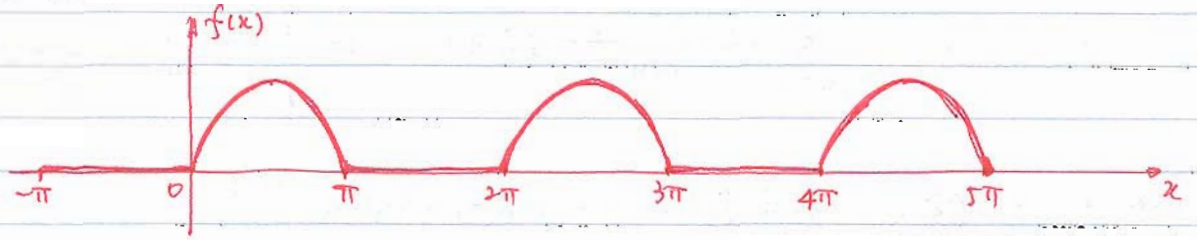
$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\int_0^{\pi}\sin x\cdot\sin nx\,dx$$

$$= \frac{1}{2\pi}\int_0^{\pi}\left[\cos(n-1)x - \cos(n+1)x\right]dx = \frac{1}{2\pi}\left[\frac{1}{n-1}\sin(n-1)x - \frac{1}{n+1}\sin(n+1)x\right]\Big|_0^{\pi} = 0,$$

$n=2,3,4,\cdots$

$$b_1 = \frac{1}{\pi}\int_0^{\pi}\sin x\cdot\sin x\,dx = \frac{1}{\pi}\int_0^{\pi}\sin^2 x\,dx$$

$$= \frac{1}{2\pi}\int_0^{\pi}(1-\cos 2x)\,dx = \frac{1}{2\pi}\left(x - \frac{1}{2}\sin 2x\right)\Big|_0^{\pi} = \frac{1}{2}.$$



∵ $f(x)$ 在 $x\in(-\infty,+\infty)$ 上连续.

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n\cos nx + b_n\sin nx)$$

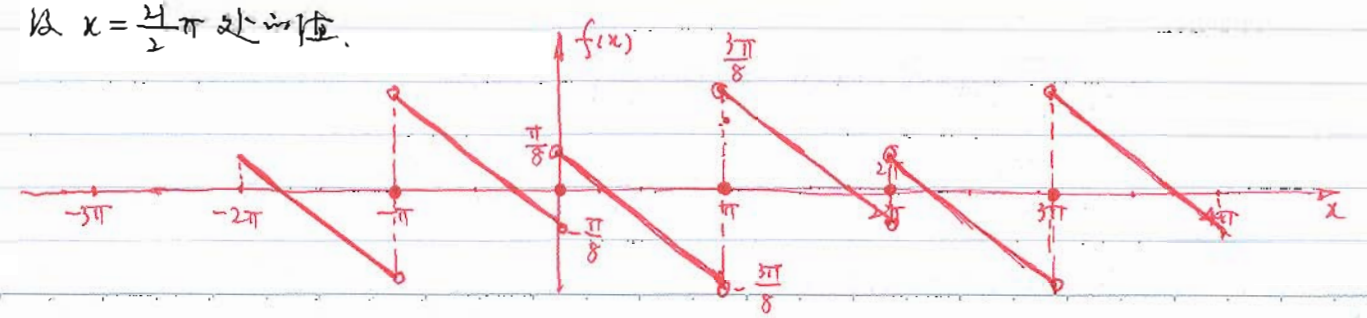
$$= \frac{1}{\pi} + \frac{1}{2}\sin x + \sum_{n=2}^{\infty}\frac{(-1)^{n+1}-1}{\pi(n^2-1)}\cos nx$$

$$= \frac{1}{\pi} + \frac{1}{2}\sin x - \sum_{n=1}^{\infty}\frac{2}{\pi(4n^2-1)}\cos 2nx$$

取 $x=\pi$, 则 $f(\pi) = \sin\pi = 0$.

$$\therefore\ 0 = \frac{1}{\pi} + \frac{1}{2}\sin\pi - \sum_{n=1}^{\infty}\frac{2}{\pi(4n^2-1)}\cdot 1 \quad\Rightarrow\quad \sum_{n=1}^{\infty}\frac{1}{1-4n^2} = -\frac{1}{2}.$$

8. 将 $f(x) = \frac{\pi}{8} - \frac{x}{2}$ $(\pi \le x < 2\pi)$ 展开成周期为 2π 的正弦级数, 并求和函数在 $x=10\pi$ 及 $x=\frac{11}{2}\pi$ 处的值.



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将 $f(x)$ 作延拓: $f(x)=\begin{cases}\frac{\pi}{8}-\frac{x}{2} & x\in[0,\pi)\\ \frac{7\pi}{8}-\frac{x}{2} & x\in[-\pi,2\pi)\end{cases}$ 或 $f(x)=\begin{cases}-\frac{\pi}{8}-\frac{x}{2} & x\in[-\pi,0)\\ \frac{\pi}{8}-\frac{x}{2} & x\in[0,\pi)\end{cases}$

延拓函数周期为 2π, 且都是奇函数.

$a_n=0, \quad n=0,1,2,\cdots\cdots$

$$b_n=\frac{2}{\pi}\int_0^{\pi}f(x)\sin nx\,dx=\frac{2}{\pi}\int_0^{\pi}\left(\frac{\pi}{8}-\frac{x}{2}\right)\sin nx\,dx$$

$$=\frac{2}{n\pi}\int_0^{\pi}\left(\frac{x}{2}-\frac{\pi}{8}\right)d\cos nx=\frac{2}{n\pi}\left(\frac{x}{2}-\frac{\pi}{8}\right)\cos nx\Big|_0^{\pi}-\frac{1}{n\pi}\int_0^{\pi}\cos nx\,dx$$

$$=\frac{1}{4n}\left[1+3(-1)^n\right] \qquad n=1,2,3,\cdots$$

$$\therefore\ f(x)\sim\sum_{n=1}^{\infty}\frac{1+3\cdot(-1)^n}{4n}\sin nx=\begin{cases}0 & x=-\pi\\ -\frac{\pi}{8}-\frac{x}{2} & -\pi<x<0\\ 0 & x=0\\ \frac{\pi}{8}-\frac{x}{2} & 0<x<\pi\end{cases}.$$

在 $x=10\pi$ 处: 和函数 $S(x)=S(10\pi)=S(0)=0$.

在 $x=\frac{21}{2}\pi$ 处: $S(x)=S\left(\frac{21}{2}\pi\right)=S\left(\frac{\pi}{2}\right)=\frac{\pi}{8}-\frac{\pi}{4}=-\frac{\pi}{8}$.

9. 设函数 f 以 2π 为周期, 且在 $[-\pi,\pi]$ 上连续, 证明:

(1) 若在 $[-\pi,\pi]$ 上成立 $f(x)=f(x+\pi)$, 则 $a_{2n+1}=b_{2n+1}=0$.

(2) 若在 $[-\pi,\pi]$ 上成立 $f(x)=-f(x+\pi)$, 则 $a_{2n}=b_{2n}=0$.

证明: (1) $a_{2n+1}=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\cos(2n+1)x\,dx=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x+\pi)\cos(2n+1)x\,dx$. $\left(令\ x+\pi=t\right)$

$$=\frac{1}{\pi}\int_0^{2\pi}f(t)\cos(2n+1)(t-\pi)\,dt=\frac{1}{\pi}\int_0^{2\pi}f(t)\cdot\cos\left[(2n+1)t+\pi\right]dt$$

$$=-\frac{1}{\pi}\int_0^{2\pi}f(t)\cdot\cos(2n+1)t\,dt=-\frac{1}{\pi}\int_{-\pi}^{\pi}f(t)\cos(2n+1)t\,dt=-a_{2n+1}.$$

$$b_{2n+1}=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\sin(2n+1)x\,dx=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x+\pi)\cdot\sin(2n+1)x\,dx.\ \left(令\ x+\pi=t\right)$$

$$=\frac{1}{\pi}\int_0^{2\pi}f(t)\sin(2n+1)(t-\pi)\,dt=\frac{1}{\pi}\int_0^{2\pi}f(t)\cdot\sin\left[(2n+1)t+\pi\right]dt$$

$$=-\frac{1}{\pi}\int_0^{2\pi}f(t)\cdot\sin(2n+1)t\,dt=-\frac{1}{\pi}\int_{-\pi}^{\pi}f(t)\sin(2n+1)t\,dt=-b_{2n+1}.$$

$\therefore\ a_{2n+1}=b_{2n+1}=0.$

(f 以 2π 为周期, $\cos 2nx$, $\sin 2nx$ 也以 2π 为周期

即 $f(x)\cdot\cos(2n+1)x$ 和 $f(x)\cdot\sin(2n+1)x$ 也以 2π 为周期

$\Rightarrow\int_0^{2\pi}=\int_{-\pi}^{\pi}$)

(2) $a_{2n}=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos 2nx\,dx = -\frac{1}{\pi}\int_{-\pi}^{\pi} f(x+\pi)\cos 2nx\,dx$. $(令\ x+\pi=t)$.

$=-\frac{1}{\pi}\int_{0}^{2\pi} f(t)\cdot\cos 2n(t-\pi)\,dt = -\frac{1}{\pi}\int_{0}^{2\pi} f(t)\cdot\cos 2nt\,dt$

$=-\frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos 2nt\,dt = -a_{2n}$.

$b_{2n}=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin 2nx\,dx = -\frac{1}{\pi}\int_{-\pi}^{\pi} f(x+\pi)\sin 2nx\,dx$. $(令\ x+\pi=t)$.

$=-\frac{1}{\pi}\int_{0}^{2\pi} f(t)\cdot\sin 2n(t-\pi)\,dt = -\frac{1}{\pi}\int_{0}^{2\pi} f(t)\cdot\sin 2nt\,dt$

$=-\frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cdot\sin 2nt\,dt = -b_{2n}$.

$\therefore\ a_{2n}=b_{2n}=0$.

或: $a_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_{-\pi}^{0} f(x)\cos nx\,dx + \frac{1}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx$

而 $\frac{1}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_{-\pi}^{0} f(t+\pi)\cos n(t+\pi)\,dt$

$=\frac{1}{\pi}\int_{-\pi}^{0}(-1)^n f(x+\pi)\cos nx\,dx$

$\therefore\ a_n=\frac{1}{\pi}\int_{-\pi}^{0}\left[f(x)+(-1)^n f(x+\pi)\right]\cos nx\,dx$

同理: $b_n=\frac{1}{\pi}\int_{-\pi}^{0}\left[f(x)+(-1)^n f(x+\pi)\right]\sin nx\,dx$.

$\therefore$ 当 $f(x)=f(x+\pi)$ 时: $a_{2n-1}=b_{2n-1}=0$.

当 $f(x)=-f(x+\pi)$ 时: $a_{2n}=b_{2n}=0$.

10. 设函数 f 在 $(0,\frac{\pi}{2})$ 上连续且有界，分别对它进行怎样的延拓，才能使它在 $[-\pi,\pi]$ 上的 Fourier 级数分别形式为:

(1) $f(x)\sim\sum_{n=1}^{\infty} a_n\cos(2n-1)x$.

解: 上述 Fourier 级数中，$b_n=0 \Rightarrow f(x)$ 为偶函数，即 $f(x)=f(-x)$.

且 $a_{2n}=0$. 即:

$a_{2n}=\frac{2}{\pi}\int_{0}^{\pi} f(x)\cos 2nx\,dx = \frac{2}{\pi}\int_{0}^{\frac{\pi}{2}} f(x)\cdot\cos 2nx\,dx + \frac{2}{\pi}\int_{\frac{\pi}{2}}^{\pi} f(x)\cos 2nx\,dx$

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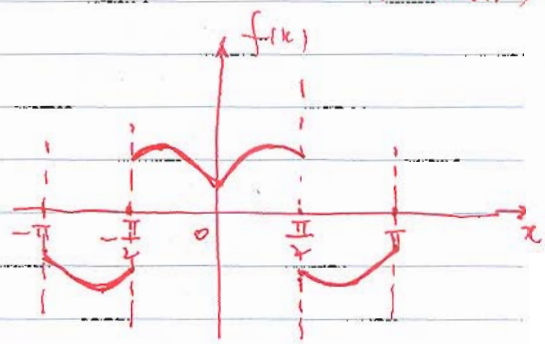
Date

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} f(x)\cos 2nx\,dx + \frac{2}{\pi}\int_0^{\frac{\pi}{2}} f(\pi-t)\cos 2nt\,dt. \quad (t=\pi-x)$$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} [f(x)+f(\pi-x)]\cos 2nx\,dx = 0. \quad \therefore f(x)+f(\pi-x)=0. \quad (\frac{\pi}{2}\text{点中心对称})$$

则 $f(x)$ 可按以下方式延拓：

$$\widetilde{f}(x) = \begin{cases} -f(\pi+x) & x\in(-\pi,-\frac{\pi}{2}) \\ f(-x) & x\in(-\frac{\pi}{2},0) \\ f(x) & x\in(0,\frac{\pi}{2}) \\ -f(\pi-x) & x\in(\frac{\pi}{2},\pi) \end{cases}$$



(2) $f(x)\sim\sum_{n=1}^{\infty} b_n \sin 2nx$.

解：由 Fourier 级数中 $a_n=0$. $\Rightarrow$ $f(x)$ 为奇函数. $f(x)=-f(-x)$. (0点中心对称)

且 $b_{2n-1}=0$. 即：

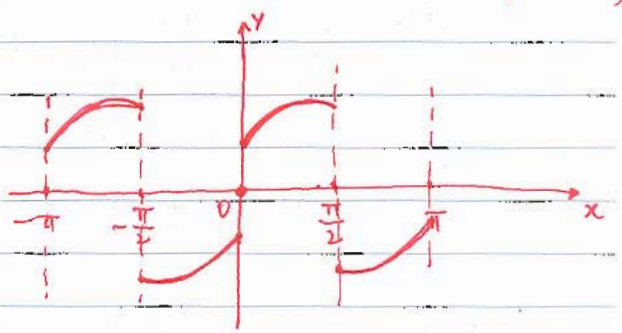
$$b_{2n-1} = \frac{2}{\pi}\int_0^{\pi} f(x)\sin(2n-1)x\,dx = \frac{2}{\pi}\int_0^{\frac{\pi}{2}} f(x)\sin(2n-1)x\,dx + \frac{2}{\pi}\int_{\frac{\pi}{2}}^{\pi} f(x)\sin(2n-1)x\,dx$$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} f(x)\sin(2n-1)x\,dx + \frac{2}{\pi}\int_0^{\frac{\pi}{2}} f(\pi-t)\sin(2n-1)t\,dt \quad (t=\pi-x)$$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} [f(x)+f(\pi-x)]\sin(2n-1)x\,dx = 0. \quad \therefore f(x)+f(\pi-x)=0. \quad (\frac{\pi}{2}\text{点中心对称})$$

则 $f(x)$ 可按以下方式延拓：

$$\widetilde{f}(x) = \begin{cases} f(\pi+x) & x\in(-\pi,-\frac{\pi}{2}) \\ -f(-x) & x\in(-\frac{\pi}{2},0) \\ f(x) & x\in(0,\frac{\pi}{2}) \\ -f(\pi-x) & x\in(\frac{\pi}{2},\pi) \end{cases}$$



11. 设函数 f 在 $[-\pi,\pi]$ 上的 Fourier 系数为 a_n 和 b_n. 求下列函数的 Fourier 系数 $\widetilde{a}_n$ 和 $\widetilde{b}_n$.

(1) $g(x)=f(-x)$.

解：$\widetilde{a}_n = \frac{1}{\pi}\int_{-\pi}^{\pi} g(x)\cos nx\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi} f(-x)\cos nx\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos nt\,dt. \quad (t=-x)$

$\therefore \widetilde{a}_n = a_n. \quad n=0,1,2,\cdots$

$\widetilde{b}_n = \frac{1}{\pi}\int_{-\pi}^{\pi} g(x)\sin nx\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi} f(-x)\sin nx\,dx = -\frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin nt\,dt. \quad (t=-x)$

$\therefore \widetilde{b}_n = -b_n. \quad n=1,2,3,\cdots$

(2) $h(x)=f(x+c)$ (c为常数)

解: $\because x+c\in[-\pi,\pi]$ $\therefore x\in[-\pi-c,\pi-c]$

$$\tilde{a}_n=\frac{1}{\pi}\int_{-\pi-c}^{\pi-c}h(x)\cos nx\,dx=\frac{1}{\pi}\int_{-\pi-c}^{\pi-c}f(x+c)\cos nx\,dx \quad (令\ t=x+c)$$

$$=\frac{1}{\pi}\int_{-\pi}^{\pi}f(t)\cos n(t-c)\,dt=\frac{1}{\pi}\int_{-\pi}^{\pi}f(t)\cos nt\cos nc\,dt+\frac{1}{\pi}\int_{-\pi}^{\pi}f(t)\sin nt\sin nc\,dt$$

$$=a_n\cos nc+b_n\sin nc. \qquad n=0,1,2,\cdots$$

$$\tilde{b}_n=\frac{1}{\pi}\int_{-\pi-c}^{+\pi-c}h(x)\sin nx\,dx=\frac{1}{\pi}\int_{-\pi-c}^{\pi-c}f(x+c)\sin nx\,dx \quad (令\ t=x+c)$$

$$=\frac{1}{\pi}\int_{-\pi}^{\pi}f(t)\sin n(t-c)\,dt=\frac{1}{\pi}\int_{-\pi}^{\pi}f(t)\sin nt\cos nc\,dt-\frac{1}{\pi}\int_{-\pi}^{\pi}f(t)\cos nt\sin nc\,dt$$

$$=b_n\cos nc-a_n\sin nc. \qquad n=1,2,\cdots$$

(3) $F(x)=\frac{1}{\pi}\int_{-\pi}^{\pi}f(t)f(x-t)\,dt$ (假定积分顺序可以交换)

解: $$\tilde{a}_n=\frac{1}{\pi}\int_{-\pi}^{\pi}F(x)\cos nx\,dx=\frac{1}{\pi}\int_{-\pi}^{\pi}\left[\frac{1}{\pi}\int_{-\pi}^{\pi}f(t)f(x-t)\,dt\right]\cos nx\,dx$$

$$=\frac{1}{\pi}\int_{-\pi}^{\pi}\left[\frac{1}{\pi}\int_{-\pi}^{\pi}f(x-t)\cos nx\,dx\right]f(t)\,dt$$

$$\tilde{a}_0=\frac{1}{\pi}\int_{-\pi}^{\pi}\left[\frac{1}{\pi}\int_{-\pi}^{\pi}f(x-t)\,dx\right]f(t)\,dt=\frac{1}{\pi}\int_{-\pi}^{\pi}a_0f(t)\,dt=a_0^2$$

$$\tilde{a}_n=\frac{1}{\pi}\int_{-\pi}^{\pi}\left[a_n\cos nt-b_n\sin nt\right]f(t)\,dt \quad (利用(2)的结论)$$

$$=\frac{1}{\pi}\int_{-\pi}^{\pi}a_n\cos nt\,f(t)\,dt-\frac{1}{\pi}\int_{-\pi}^{\pi}b_n\sin nt\,f(t)\,dt=a_n^2-b_n^2 \qquad n=1,2,3\cdots$$

$$\tilde{b}_n=\frac{1}{\pi}\int_{-\pi}^{\pi}F(x)\sin nx\,dx=\frac{1}{\pi}\int_{-\pi}^{\pi}\left[\frac{1}{\pi}\int_{-\pi}^{\pi}f(t)f(x-t)\,dt\right]\sin nx\,dx$$

$$=\frac{1}{\pi}\int_{-\pi}^{\pi}\left[\frac{1}{\pi}\int_{-\pi}^{\pi}f(x-t)\sin nx\,dx\right]f(t)\,dt$$

$$=\frac{1}{\pi}\int_{-\pi}^{\pi}\left[b_n\cos nt+a_n\sin nt\right]f(t)\,dt \quad (利用(2)的结论)$$

$$=\frac{1}{\pi}\int_{-\pi}^{\pi}b_nf(t)\cos nt\,dt+\frac{1}{\pi}\int_{-\pi}^{\pi}a_nf(t)\sin nt\,dt=2a_nb_n \qquad n=1,2,3,\cdots$$

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12. 利用例 9.3.1 的结果和 Parseval 等式，重新证明等式：$\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}=\frac{\pi^2}{8}$.

证明：例 9.3.1 的结果：$f(x)\sim\frac{1}{2}+\frac{1}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n-1}{n}\sin nx$，$f(x)=\begin{cases}1 & x\in[-\pi,0)\\ 0 & x\in(0,\pi)\end{cases}$

$a_0=1$，$a_n=0$，$b_n=\frac{(-1)^n-1}{n\pi}$，$n=1,2,\cdots$

Parseval 等式：$\frac{a_0^2}{2}+\sum_{n=1}^{\infty}(a_n^2+b_n^2)=\frac{1}{\pi}\int_{-\pi}^{\pi}f^2(x)\,dx$

即：$\frac{a_0^2}{2}+\sum_{n=1}^{\infty}(a_n^2+b_n^2)=\frac{1}{2}+\sum_{n=1}^{\infty}\left[\frac{(-1)^n-1}{n\pi}\right]^2=\frac{1}{2}+\sum_{n=1}^{\infty}\frac{4}{(2n-1)^2\pi^2}$

$=\frac{1}{2}+\frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}$.

$\frac{1}{\pi}\int_{-\pi}^{\pi}f^2(x)\,dx=\frac{1}{\pi}\int_{-\pi}^{0}1\,dx=1$.

$\Rightarrow\frac{1}{2}+\frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}=1$. $\therefore\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}=\frac{\pi^2}{8}$.

13. 利用例 9.3.2 的结果和 Parseval 等式，求 $\sum_{n=1}^{\infty}\frac{1}{(2n-1)^4}$.

证明：例 9.3.2 的结果：$f(x)=\begin{cases}x & x\in[0,\pi)\\ -x & x\in[-\pi,0)\end{cases}$（偶延拓）

$a_0=\pi$，$a_n=\begin{cases}0 & n=2k\\ -\frac{4}{n^2\pi} & n=2k+1\end{cases}$，$b_n=0$，$\left(a_n=2\cdot\frac{(-1)^n-1}{n^2\pi}\right)$.

$\frac{1}{\pi}\int_{-\pi}^{\pi}f^2(x)\,dx=\frac{2}{\pi}\int_{0}^{\pi}f^2(x)\,dx=\frac{2}{\pi}\int_{0}^{\pi}x^2\,dx=\frac{2}{3}\pi^2$.

$\frac{a_0^2}{2}+\sum_{n=1}^{\infty}(a_n^2+b_n^2)=\frac{\pi^2}{2}+\sum_{n=1}^{\infty}\left[2\cdot\frac{(-1)^n-1}{n^2\pi}\right]^2=\frac{\pi^2}{2}+\sum_{n=1}^{\infty}\frac{16}{(2n-1)^4\pi^2}$

由 Parseval 等式，可得：$\frac{\pi^2}{2}+\sum_{n=1}^{\infty}\frac{16}{(2n-1)^4\pi^2}=\frac{2}{3}\pi^2$.

$\therefore\sum_{n=1}^{\infty}\frac{1}{(2n-1)^4}=\frac{\pi^2}{16}\left(\frac{2}{3}\pi^2-\frac{1}{2}\pi^2\right)=\frac{\pi^4}{96}$.

§4. Fourier 变换初步

习题. P226

1. 求下列区间 $(-\infty,+\infty)$ 内的函数的 Fourier 变换：

(1) $f(x) = e^{-a|x|}$ $(a>0)$

解: $F[f] = \int_{-\infty}^{+\infty} f(x) e^{-i\omega x} dx = \int_{-\infty}^{+\infty} e^{-a|x|} \cdot e^{-i\omega x} dx$

$= \int_{-\infty}^{0} e^{(a-i\omega)x} dx + \int_{0}^{+\infty} e^{-(a+i\omega)x} dx$

$= \frac{1}{a-i\omega} e^{(a-i\omega)x}\Big|_{-\infty}^{0} - \frac{1}{a+i\omega} e^{-(a+i\omega)x}\Big|_{0}^{+\infty}$

$= \frac{1}{a-i\omega} + \frac{1}{a+i\omega} = \frac{2a}{a^2+\omega^2}$.

(2) $f(x) = \begin{cases} e^{-2x} & x \geq 0 \\ 0 & x < 0. \end{cases}$

解: $F[f] = \int_{-\infty}^{+\infty} f(x) e^{-i\omega x} dx = \int_{-\infty}^{0} 0 \cdot e^{-i\omega x} dx + \int_{0}^{+\infty} e^{-2x} \cdot e^{-i\omega x} dx$

$= \int_{0}^{+\infty} e^{-(2+i\omega)x} dx = -\frac{1}{2+i\omega} e^{-(2+i\omega)}\Big|_{0}^{+\infty} = \frac{1}{2+i\omega}$.

(3) $f(x) = e^{-|x|}\cos x$.

解: $F[f] = \int_{-\infty}^{+\infty} f(x) e^{-i\omega x} dx = \int_{-\infty}^{0} e^{x}\cos x \cdot e^{-i\omega x} dx + \int_{0}^{+\infty} e^{-x}\cos x \cdot e^{-i\omega x} dx$

$= \int_{-\infty}^{0} e^{(1-i\omega)x}\cos x\, dx + \int_{0}^{+\infty} e^{-(1+i\omega)x}\cos x\, dx = F_1(\omega) + F_2(\omega)$.

$F_1(\omega) = \int_{-\infty}^{0} e^{(1-i\omega)x}\cos x\, dx = \int_{-\infty}^{0} e^{(1-i\omega)x} d\sin x = e^{(1-i\omega)x}\sin x\Big|_{-\infty}^{0}$

$- \int_{-\infty}^{0} \sin x \cdot d e^{(1-i\omega)x} = \frac{1}{1-i\omega} \int_{-\infty}^{0} e^{(1-i\omega)x} d\cos x$

$= \frac{1}{1-i\omega} e^{(1-i\omega)x}\cos x\Big|_{-\infty}^{0} - \frac{1}{(1-i\omega)^2} \int_{-\infty}^{0} e^{(1-i\omega)x}\cos x\, dx$

$= \frac{1}{1-i\omega} - \frac{1}{(1-i\omega)^2} F_1(\omega)$. $\therefore F_1(\omega) = \frac{\frac{1}{1-i\omega}}{1+\frac{1}{(1-i\omega)^2}} = \frac{1-i\omega}{2-\omega^2-2\omega i}$.

$e^{-\infty + i\omega\cdot\infty}$
$= e^{-\infty}(\sin\omega\cdot\infty + i\cos\omega\cdot\infty)$
$\equiv 0$

$F_2(\omega) = \int_{0}^{+\infty} e^{-(1+i\omega)x} dx = \int_{0}^{+\infty} e^{-(1+i\omega)x} d\sin x$

$= e^{-(1+i\omega)x}\sin x\Big|_{0}^{+\infty} - \int_{0}^{+\infty} \sin x\, d e^{-(1+i\omega)x} = -\frac{1}{1+i\omega} \int_{0}^{+\infty} e^{-(1+i\omega)x} d\cos x$

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$$= \frac{1}{1+i\omega} e^{-(1+i\omega)x} \cos x \Big|_0^{+\infty} - \frac{1}{(1+i\omega)^2} \int_0^{+\infty} e^{-(1+i\omega)x} \cdot \cos x \, dx$$

$$= \frac{1}{1+i\omega} - \frac{1}{(1+i\omega)^2} F_2(\omega). \quad \therefore F_2(\omega) = \frac{\frac{1}{1+i\omega}}{1+\frac{1}{(1+i\omega)^2}} = \frac{1+i\omega}{2-\omega^2+2\omega i}$$

$$\therefore F[f] = F_1(\omega) + F_2(\omega) = \frac{1-i\omega}{2-\omega^2-2\omega i} + \frac{1+i\omega}{2-\omega^2+2\omega i} = \frac{2\omega^2+4}{\omega^4+4}$$

2. 设 f 是 $(-\infty, +\infty)$ 上绝对可积的可导函数.

i) 证明 f 是偶函数时的 Fourier 积分:

$$f(x) = \frac{2}{\pi} \int_0^{+\infty} \left[\int_0^{+\infty} f(t) \cos \omega t \, dt \right] \cos \omega x \, d\omega$$

ii) 证明 f 是奇函数时的 Fourier 积分:

$$f(x) = \frac{2}{\pi} \int_0^{+\infty} \left[\int_0^{+\infty} f(t) \sin \omega t \, dt \right] \sin \omega x \, d\omega$$

证明: f 的 Fourier 积分为:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} f(x) e^{-i\omega x} dx \right] e^{i\omega x} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} f(x) (\cos \omega x - i \sin \omega x) dx \right] e^{i\omega x} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{0} f(x) (\cos \omega x - i \sin \omega x) dx + \int_0^{+\infty} f(x) (\cos \omega x - i \sin \omega x) dx \right] e^{i\omega x} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left[\int_0^{+\infty} f(-x) (\cos \omega x + i \sin \omega x) dx + \int_0^{+\infty} f(x) (\cos \omega x - i \sin \omega x) dx \right] e^{i\omega x} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left[\int_0^{+\infty} (f(-x) + f(x)) \cos \omega x \, dx + i \int_0^{+\infty} (f(-x) - f(x)) \sin \omega x \, dx \right] e^{i\omega x} d\omega$$

i) 当 f 是偶函数时, $f(x) = f(-x)$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left[\int_0^{+\infty} 2 f(x) \cos \omega x \, dx \right] e^{i\omega x} d\omega$$

$$= \frac{1}{\pi} \left\{ \int_{-\infty}^{0} \left[\int_0^{+\infty} f(x) \cos \omega x \, dx \right] (\cos \omega x + i \sin \omega x) \, d\omega \right.$$

$$\left. + \int_0^{+\infty} \left[\int_0^{+\infty} f(x) \cos \omega x \, dx \right] (\cos \omega x + i \sin \omega x) d\omega \right\}$$

$$= \frac{1}{\pi}\left\{ \int_0^{+\infty}\left[\int_0^{+\infty} f(x)\cos\omega x\,dx\right](\cos\omega x - i\sin\omega x)\,d\omega \right.$$
$$\left. + \int_0^{+\infty}\left[\int_0^{+\infty} f(x)\cos\omega x\,dx\right](\cos\omega x + i\sin\omega x)\,d\omega \right\}$$
$$= \frac{1}{\pi}\left\{ 2\int_0^{+\infty}\left[\int_0^{+\infty} f(x)\cos\omega x\,dx\right]\cos\omega x\,d\omega \right\}$$
$$= \frac{2}{\pi}\int_0^{+\infty}\left[\int_0^{+\infty} f(t)\cos\omega t\,dt\right]\cos\omega x\,d\omega$$

(2) 当 f 为奇函数时, $f(x) = -f(-x)$.

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{+\infty}\left[-2i\int_0^{+\infty} f(x)\sin\omega x\,dx\right]e^{i\omega x}\,d\omega$$
$$= -\frac{i}{\pi}\left\{ \int_{-\infty}^{0}\left[\int_0^{+\infty} f(x)\sin\omega x\,dx\right](\cos\omega x + i\sin\omega x)\,d\omega \right.$$
$$\left. + \int_0^{+\infty}\left[\int_0^{+\infty} f(x)\sin\omega x\,dx\right](\cos\omega x + i\sin\omega x)\,d\omega \right\}$$
$$= -\frac{i}{\pi}\left\{ \int_0^{+\infty}\left[-\int_0^{+\infty} f(x)\sin\omega x\,dx\right](\cos\omega x - i\sin\omega x)\,d\omega \right.$$
$$\left. + \int_0^{+\infty}\left[\int_0^{+\infty} f(x)\sin\omega x\,dx\right](\cos\omega x + i\sin\omega x)\,d\omega \right\}$$
$$= -\frac{i}{\pi}\left\{ 2i\int_0^{+\infty}\left[\int_0^{+\infty} f(x)\sin\omega x\,dx\right]\sin\omega x\,d\omega \right\}$$
$$= \frac{2}{\pi}\int_0^{+\infty}\left[\int_0^{+\infty} f(t)\sin\omega t\,dt\right]\sin\omega x\,d\omega$$

3. 利用 1(1) 和 2(1) 证明:

(1) $\displaystyle\int_0^{+\infty}\frac{\cos\omega x}{a^2+\omega^2}\,d\omega = \frac{\pi}{2a}e^{-a|x|}$.

证明: 由 1(1) 可知: $f(x) = e^{-a|x|}$ $(a>0)$. 有: $F[f] = \hat{f}(\omega) = \dfrac{2a}{a^2+\omega^2}$.

且 $f(x)$ 在 $(-\infty,+\infty)$ 连续. 即有: $f(x) = \dfrac{1}{2\pi}\displaystyle\int_{-\infty}^{+\infty}\hat{f}(\omega)e^{i\omega x}\,d\omega$

即: $e^{-a|x|} = \dfrac{1}{2\pi}\displaystyle\int_{-\infty}^{+\infty}\hat{f}(\omega)e^{i\omega x}\,d\omega = \frac{1}{2\pi}\left(\int_{-\infty}^{0}+\int_0^{+\infty}\right)\frac{2a}{a^2+\omega^2}e^{i\omega x}\,d\omega$

$$= \frac{1}{2\pi}\left[\int_0^{+\infty}\frac{2a}{a^2+\omega^2}e^{-i\omega x}\,d\omega + \int_0^{+\infty}\frac{2a}{a^2+\omega^2}e^{i\omega x}\,d\omega\right]$$

$$= \frac{1}{2\pi}\left[\int_0^{+\infty}\frac{2a}{a^2+\omega^2}\cdot 2\cos\omega x\,d\omega\right] = \frac{2a}{\pi}\int_0^{+\infty}\frac{\cos\omega x}{a^2+\omega^2}\,d\omega.$$

$$\therefore \int_0^{+\infty}\frac{\cos\omega x}{a^2+\omega^2}\,d\omega = \frac{\pi}{2a}e^{-a|x|}.$$

(2) 设 φ 是 $[0,+\infty)$ 上绝对可积的函数，$\int_0^{+\infty}\varphi(y)\cos(xy)\,dy = \frac{1}{1+x^2}$ $(x\geq 0)$. 求 $\varphi(x)$.

解：在(1)中，令 $a=1$，则有：

$$e^{-|x|} = \frac{2}{\pi}\int_0^{+\infty}\frac{\cos\omega x}{1+\omega^2}\,d\omega. \quad \text{即} \quad \frac{1}{1+\omega^2} = \int_0^{+\infty}\varphi(x)\cos\omega x\,dx. \quad (\text{全条件中 } x=\omega)$$

则：$e^{-|x|} = \frac{2}{\pi}\int_0^{+\infty}\left[\int_0^{+\infty}\varphi(x)\cos\omega x\,dx\right]\cos\omega x\,d\omega = \varphi(x)$. (由2(1)结论)

(若 $(-\infty,+\infty)$ 上作偶延拓)

$$\therefore \varphi(x) = e^{-x}, \quad x\in[0,+\infty).$$

4. 证明离散的正交关系式：$\frac{1}{N}\sum_{n=0}^{N-1}e^{-2\pi i\frac{nk}{N}}e^{2\pi i\frac{nj}{N}} = \delta_{j,k}$

证明：当 $j=k$ 时，显然有：

$$\frac{1}{N}\sum_{n=0}^{N-1}e^{-2\pi i\frac{nk}{N}}e^{2\pi i\frac{nj}{N}} = \frac{1}{N}\sum_{n=0}^{N-1}e^{-2\pi i\cdot\frac{n}{N}(k-j)} = \frac{1}{N}\sum_{n=0}^{N-1}e^{0} = \frac{1}{N}\sum_{n=0}^{N-1}1 = 1.$$

当 $j\neq k$ 时，不妨设 $j<k$.

若 ξ $(\neq 1)$ 是方程 $x^N=1$ 的一个根时，有 $\sum_{n=0}^{N-1}\xi^n = 0$.

$[\xi^N-1 = (\xi-1)(\xi^{N-1}+\xi^{N-2}+\cdots+\xi+1) = 0$. 即得 $\sum_{n=0}^{N-1}\xi^n = 0$ $(\xi\neq 1)]$

令 $\xi = e^{2\pi i\frac{(k-j)}{N}} \neq 1$. 则 $\xi^N = e^{2\pi i(k-j)} = \cos(k-j)2\pi + i\sin(k-j)2\pi = 1$.

$$\therefore \sum_{n=0}^{N-1}\xi^n = \sum_{n=0}^{N-1}\left(e^{2\pi i\cdot\frac{(k-j)}{N}}\right)^n = \sum_{n=0}^{N-1}e^{2\pi i\frac{nk}{N}}\cdot e^{-2\pi i\frac{nj}{N}} = 0.$$

$$\therefore \frac{1}{N}\sum_{n=0}^{N-1}e^{-2\pi i\frac{nk}{N}}e^{2\pi i\frac{nj}{N}} = \delta_{j,k}.$$